

1. SAP Data Services: User Guide	2
1.1 SAP Data Services: Change Log	4
1.2 SAP Data Services: Service Description & Architecture	5
1.2.1 SAP Data Services: Service Description	6
1.2.2 SAP Data Services: Architecture Overview	7
1.2.3 SAP Data Services: Environment types	10
1.2.4 SAP Data Services: Environment Setups	11
1.2.5 SAP Data Services: Cloud Connectivity	13
1.2.6 SAP Data Services: How to connect Designer to COP	15
1.3 SAP Data Services: Getting Started	19
1.3.1 SAP Data Services: What requests can you make?	20
1.3.2 SAP Data Services: How to request the creation or modification of an instance	22
1.4 SAP Data Services: Learn about your service	23
1.4.1 SAP Data Services: Backup & restore	24
1.4.2 SAP Data Services: Upgrading the service	26
1.4.3 SAP Data Services: Versions & Patches	27
1.4.4 SAP Data Services: Licenses	28
1.4.5 SAP Data Services: How to decommission the service	29
1.5 SAP Data Services: Authentication	30
1.6 SAP Data Services: Tools	32
1.6.1 SAP Data Services: Designer tool	33
1.6.2 SAP Data Services: Extract-Transform-Load (ETL) tool	34
1.6.3 SAP Data Services: Management Console	36
1.7 SAP Data Services: Repository & Staging Areas	37
1.7.1 SAP Data Services: SAP Data Services repository	38
1.7.2 SAP Data Services: Staging & Sharing Areas	41
1.8 SAP Data Services: Monitoring	42
1.9 SAP Data Services: FAQ & Supporting Links	43
1.9.1 SAP Data Services: FAQ	44
1.9.1.1 SAP Data Services: General FAQ	45
1.9.1.2 SAP Data Services: General Recommendations	47
1.9.1.3 SAP Data Services: How to connect EUSend with SAP Data Services	49
1.9.1.4 SAP Data Services: How to connect SUMMA DWH with SAP Data Services	60
1.9.1.5 SAP Data Services: How to use PrivX with SAP Data Services	69
1.9.2 SAP Data Services: Support & Useful Links	74

SAP Data Services: User Guide



Change Log



Check the **Change Log** to see what's new and track past changes

Service Description & Architecture



[Service Description](#)
[Architecture Overview](#)
[Environment types](#)
[Environment Setups](#)
[Cloud Connectivity](#)
[How to connect Designer to COP](#)

Getting Started

GET STARTED



What requests can you make?

Request the creation or modification of an instance

Learn about your service



[Backup & restore](#)
[Upgrading the service](#)
[Versions & Patches](#)
[Licenses](#)
[How to decommission the service](#)

Securing your service



[Authentication](#)

Tools



[Management Console](#)
[Designer tool](#)
[Extract-Transform-Load \(ETL\) tool](#)

Repository & Staging Areas



[Staging & Sharing Areas](#)

[SAP Data Services repository](#)

Monitoring



[Monitoring](#)

FAQ & Supporting Links



[General FAQ](#)

[General Recommendations](#)

[How to connect EUSend with SAP Data Services](#)

[How to connect SUMMA DWH with SAP Data Services](#)

[How to use PrivX with SAP Data Services](#)

[Support & Useful Links](#)

SAP Data Services: Change Log

Date	Change Description	Link
01 Sep 2025	How to use connect to SAP DS repositories on Oracle COP	https://webgate.ec.europa.eu/fpfis/wikis/x/7Iplgw

SAP Data Services: Service Description & Architecture

- [SAP Data Services: Service Description](#)
- [SAP Data Services: Architecture Overview](#)
- [SAP Data Services: Environment types](#)
- [SAP Data Services: Environment Setups](#)
- [SAP Data Services: Cloud Connectivity](#)
- [SAP Data Services: How to connect Designer to COP](#)

SAP Data Services: Service Description



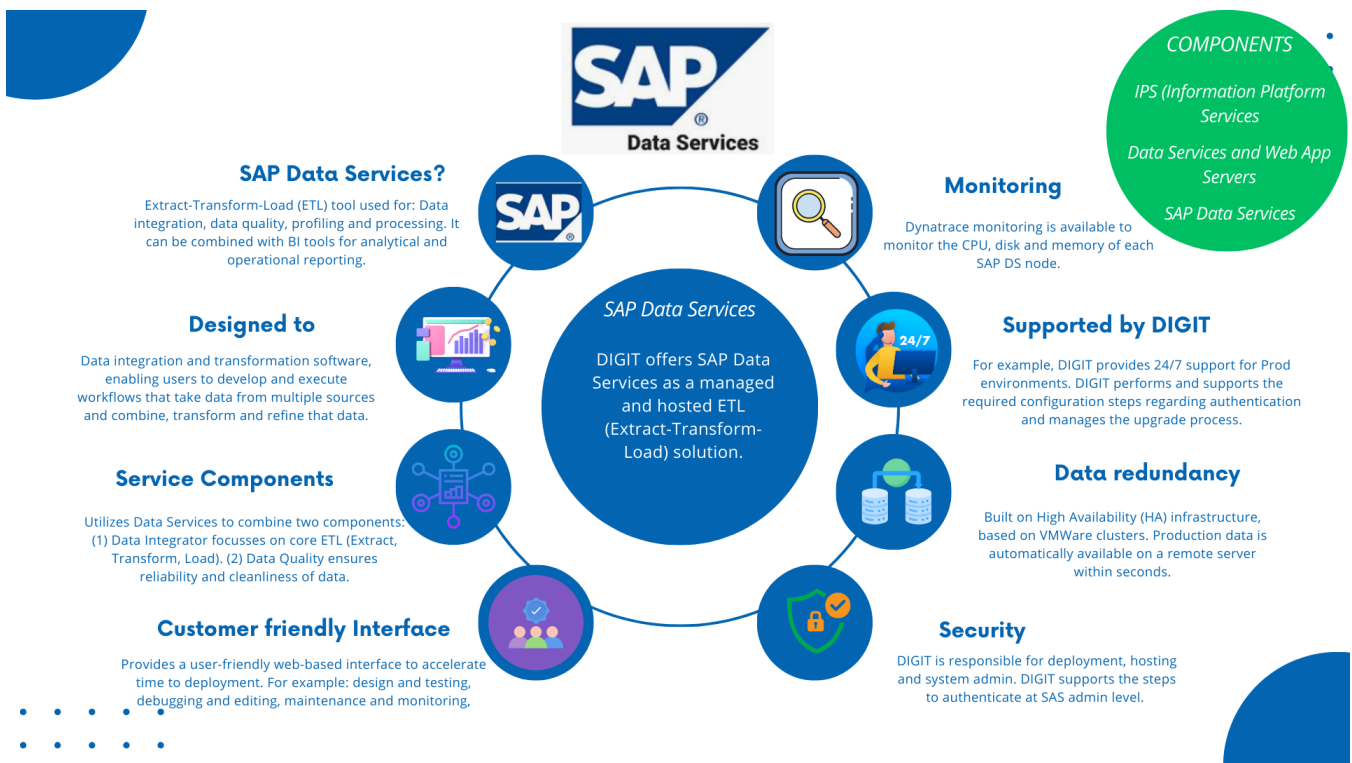
Homepage

Related Links

[Architecture Overview](#)

[Environment types](#)

[Environment Setups](#)



SAP Data Services: Architecture Overview



Homepage

Page Topics

- [What is the deployment architecture?](#)
- [What does the architecture comprise?](#)
- [What comprises the Production architecture?](#)
 - [High Availability](#)
- [What comprises the Development and Test architecture?](#)
- [Can a setup be customised?](#)

Related Links

- [Service Description](#)
- [Environment types](#)
- [Environment Setups](#)

What is the deployment architecture?

- One stand-alone virtual machine for Development and Test environments.
- Two dedicated Virtual Machines (VMs) for Production environments.

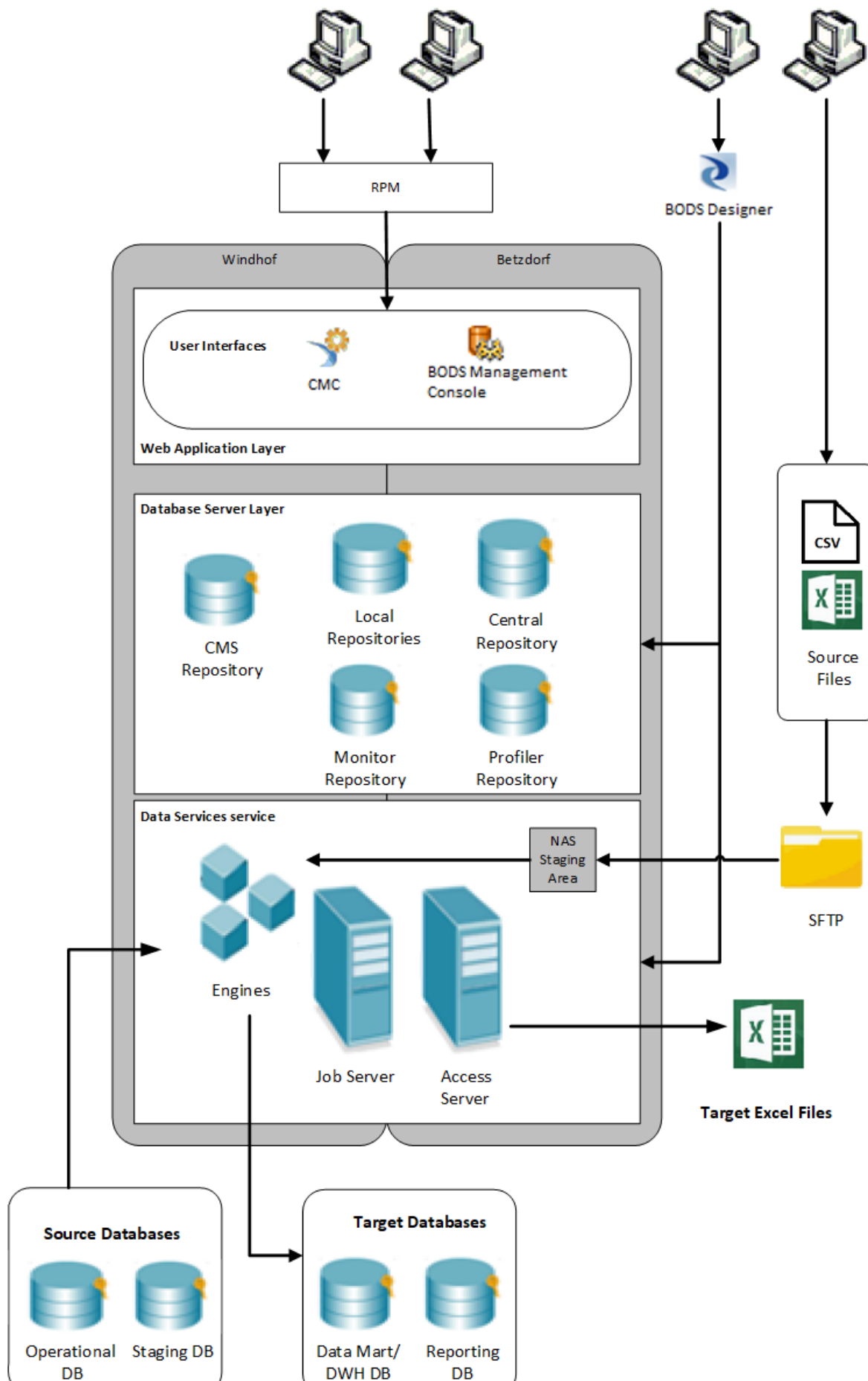
What does the architecture comprise?

Each SAP Data Service instance:

- Is deployed on one (Stand-alone) Virtual Machine (VM) or a cluster of VMs (Clustered).
- Runs on the Red Hat Enterprise Linux operating system.

A SAP Data Service instance is made of three main components:

- IPS (Information Platform Services)
- Data Services
- Web Application servers (The default web application server is Tomcat)





What comprises the Production architecture?

- Each SAP Data Services instance is deployed using two dedicated Virtual Machines (VMs)
- The sizing of the VM is based on the ETL (Extract-Transform-Load) expected workload and DIGIT license compliance.
- Environments are supported 24/7

High Availability

- Runs on a pair of Virtual Machines (VMs) split across datacentres.
- If a server or a datacentre becomes unavailable, high availability is maintained by running on one leg of the architecture (Using a Load Balancer and Virtual Machine (VM) pairing).

What comprises the Development and Test architecture?

- Each SAP Data Services instance is deployed on one single Virtual Machine (VM)
- No fail-over mechanism is available
- Environments are supported 12/5
- The sizing of the VM is based on the ETL (Extract-Transform-Load) expected workload and DIGIT license compliance.

Can a setup be customised?

- Custom setups are possible and are evaluated by DIGIT on a case-by-case basis.
- The final decision to support a customised instance rests with DIGIT.

SAP Data Services: Environment types



Homepage

ENVIRONMENT TYPES		
ENVIRONMENT	ABBREVIATION	DESCRIPTION
Development	DEV	• Development purposes. • Administration access can also be provided upon request. • No fail-over mechanism. The environment will be unavailable until the problem is solved.
Test	TST	• Testing the service before it goes into production. • No fail-over mechanism. The environment will be unavailable until the problem is solved.
Production	PRD	HIGH AVAILABILITY • High availability is maintained using the combination of Load Balancer and Virtual Machine (VM) pairing. DATA PROTECTION • Production IT services and systems are eligible to be classified as critical or essential services and have priority attention in cases of invoked business continuity (BCP) and disaster recovery (DRP) plans. SUPPORT • Supported 24x7

Related Links

- [Service Description](#)
- [Architecture Overview](#)
- [Environment Setups](#)

SAP Data Services: Environment Setups



Homepage

Page Topics

- [Can environment settings be modified?](#)
- [How to modify an environment](#)
- [What size are environments?](#)
- [Who can access an environment?](#)
- [How to create an IS/environment in the datacentre](#)

Related Links

- [Service Description](#)
- [Architecture Overview](#)
- [Environment types](#)

Can environment settings be modified?

If specified by the customer, additional settings can be applied (if not in contradiction of any DIGIT standards)

How to modify an environment

1. Make an [RFC](#) or [RFS](#) in [JASSPR](#) (Modify SAP Data Service Instance)
 - a. Complete the fields in the form
 - b. Include as much useful information as possible.
 - c. Submit the form.
 - d. Monitor the request status on the [My requests list](#).
2. For information and examples how to use [JASSPR](#) - consult the [JASSPR User Guide](#)

What size are environments?

The sizing of the VM is based on the ETL (Extract-Transform-Load) expected workload and DIGIT license compliance.

Who can access an environment?

You must have access to the following:

- Commission, Executive Agencies and/or [TESTA](#) networks.
- Either WELCOME, NET1 or RCNET domain to be able to use this service.

How to create an IS/environment in the datacentre

Watch the following video to learn how to quickly create an Information System (IS) and environment in the datacentre.

You can also download the video [here](#). You will use [JASSPR](#) to do achieve this.

Your browser does not support the HTML5 video element



For more information about JASSPR, refer to the [User Guide](#) and watch the JASSPR tutorial videos [here](#).

SAP Data Services: Cloud Connectivity



Page Topics

- [Where to Find SAP Cloud Connectivity Compatibility](#)
- [What kind of cloud connections are supported by SAP DS 4.3?](#)
- [Use of WS to connect to cloud data sources](#)

Related Links

- [Service Description](#)
- [Environment types](#)
- [Architecture Overview](#)
- [Environment Setups](#)

Where to Find SAP Cloud Connectivity Compatibility

For **SAP Data Services 4.3**, compatibility information regarding **cloud connectivity** can be found in the official **Product Availability Matrix (PAM)**.

Refer to the following SAP Knowledge Base Article:
[1338845 - How to find Product Availability Matrix \(PAM\) / Supported Platforms Documentation for SAP BusinessObjects products](#)

In PAM, you will find detailed information on:

- Supported cloud platforms and environments
- Connectivity options available for SAP Data Services 4.3

In addition to native support, **alternative integration methods** such as **Web Services** or **Python scripts** can also be used as workarounds to enable cloud connectivity with SAP Data Services.

What kind of cloud connections are supported by SAP DS 4.3?

Below, the cloud connections supported by SAP DS 4.3 are displayed:

Cloud Database	Version	Client	Windows	Linux
			Source, Target	Source, Target
SAP HANA Data Lake Database	QRC 1/2023	SAP HANA Cloud, datalake Relational Engine 17.01.06	SP01	SP01
	QRC 1/2023	SAP HANA Cloud, datalake Relational Engine 17.1.232.2935	SP03	SP03
SAP DataSphere/ Data Warehouse Cloud (DWC)	2023.11	HANA ODBC 2.12.13 HANA ODBC 2.16.26 ⁽¹⁾ HANA ODBC 2.19.20 ⁽²⁾	✓	✓
SAP HANA Cloud	2023.04 (QRC 1/2023)	HANA ODBC 2.12.13 HANA ODBC 2.16.26 ⁽¹⁾ HANA ODBC 2.19.20 ⁽²⁾	✓	✓
Google BigQuery	Current version	ODBC Driver 2.4.5.1014 for Big Query	✓	✓
		ODBC release 2.5.2.1004 ODBC release 3.0.0.1001 ODBC release 3.0.4.1008	SP03	SP03
Snowflake	Current version	Snowflake ODBC Driver 2.24.0 Snowflake ODBC Driver 2.25.11	✓	✓
		Snowflake ODBC Driver 3.1.4	SP03	SP03

Amazon Redshift	Current version	64-bit ODBC driver version 1.5.7	✓	✓
		64-bit ODBC driver version 2.1.1.0	SP03 Patch 1	SP03 Patch 1
Amazon Athena	Current version	ODBC Driver 1.1.19.1002	SP02	SP02
		ODBC Driver 1.1.20.1002		
		ODBC Driver 1.1.20.1003	SP03	SP03
Amazon RDS for SQL Server	2019	Microsoft ODBC Driver 17 for SQL Server (Windows) DataDirect ODBC Driver V8.0 SP2 (Linux)	✓	SP00
	2022	Microsoft ODBC Driver 17 for SQL Server (Windows) Microsoft ODBC Driver 18 for SQL Server (Windows) DataDirect ODBC Driver V8.0 SP2 (Linux)	SP03 Patch 1	SP03 Patch 1
MS SQL Server on Azure VM	All SQL server supported versions	Microsoft ODBC Driver 17 for SQL Server (Windows) Microsoft ODBC Driver 18 for SQL Server (Windows) DataDirect ODBC Driver V8.0 SP2 (Linux)	✓	✓
Azure SQL Database	Current version	Microsoft ODBC Driver 17 for SQL Server (Windows) Microsoft ODBC Driver 18 for SQL Server (Windows) DataDirect ODBC Driver V8.0 SP2 (Linux)	✓	✓
Azure Synapse	Current version	Microsoft ODBC Driver 17 for SQL Server (Windows) Microsoft ODBC Driver 18 for SQL Server (Windows) DataDirect ODBC Driver V8.0 SP2 (Linux)	✓	✓

(1) HANA ODBC 2.16.26 is supported since Data Services 4.3 SP02 Patch 2.

(2) HANA ODBC 2.19.20 is supported since Data Services 4.3 SP03.

It's also recommended to upgrade to SAP DataServices 2025 as it will come with more connectivity possibilities.

Use of WS to connect to cloud data sources

- SAP DataServices supports the use of WebServices as a workaround which can be used with SAP DataStores.
- DIGIT recommends that you activate SSL for the WS: [1771712 - How to configure SSL for a Web Service Datastore - SAP Data Services and SAP Cloud Platform Integration for Data Services | SAP Knowledge Base Article](#)

SAP Data Services: How to connect Designer to COP



Page Topics

- [Introduction](#)
- [Pre-requisites](#)
- [Connecting to COP DB \(PDB\) from PC / Application](#)
 - [Connect to PRIVX VPN](#)
 - [Connect to PAM Privx](#)
 - [Connect with a client tool](#)
- [List of Network Targets for SAP DS](#)

Related Links

- [Service Description](#)
- [Environment types](#)
- [Architecture Overview](#)
- [Environment Setups](#)

Introduction

With this document we want to explain step by step to connect your client tools (SAP BI and SAP DS Designer) to COP.

Pre-requisites

To be able to connect to a COP DB Source used by SAP BI, a repository or a DB used by SAP DS we need to have the below pre-requisites:

- Have installed to PAM PrivX Gateway (available in EC Store):

- Have Access to PrivX Network Targets by PAM.
- Have access to a target network used by the COP DB.
- Have a client tool to use it: SAP BO BI or SAP DS.

Connecting to COP DB (PDB) from PC / Application

Connect to PRIVX VPN

We need to go to Settings, VPN and then select PAM Privx Gateway:

COSMOS > SAP BI/DS Client Tools Connect to COP > image-2025-3-5_16-38-23.
png

Connect to PAM Privx

Go to <https://pam.tech.ec.europa.eu/> and follow the steps (Connections → Network Targets):

COSMOS > SAP BI/DS Client Tools Connect to COP > image-2025-3-5_16-39-20.
png

When connected select the network to connect where the DB is located:

COSMOS > SAP BI/DS Client Tools Connect to COP > image-2025-3-5_16-40-8.
png

Connect with a client tool

Now we can open an SQL-Developer tool and try the connection or just open the SAP BI Client Tools or SAP DS Designer to use as normal:

COSMOS > SAP BI/DS Client Tools Connect to COP > image-2025-3-5_16-40-49.
png

LDAP: [oid.cc.cec.eu.int:389](ldap://oid.cc.cec.eu.int:389)

Context: cn=OracleContext,dc=prod,dc=cop,dc=cec,dc=eu,dc=int

List of Network Targets for SAP DS

Here are the target networks used for SAP DS repositories (LOCAL, PROFILER and CENTRAL):

COSMOS > SAP BI/DS Client Tools Connect to COP > image-2025-11-11_15-26-40.
png

- **BIDSOSI_REP_NON-PROD** for non productive environments.
- **BIDSOSI_REP_PROD** for productive environments.

SAP Data Services: Getting Started

- [SAP Data Services: What requests can you make?](#)
- [SAP Data Services: How to request the creation or modification of an instance](#)

SAP Data Services: What requests can you make?



Homepage

Page Topics

- [What are RFS and RFC?](#)
- [How to make an RFC / RFS request](#)
- [What is an RFI?](#)
- [How to make an RFI request](#)
- [How to find more information](#)

Related Links

[Request the creation or modification of an instance](#)

RFC		
Create	Modify	Remove
• SAP Data Services Instance	• SAP Data Services Instance (Add Node) • SAP Data Services Instance (Remove Node)	• Decommission SAP Data Services Instance

What are RFS and RFC?

Request for Service ([RFS](#))

- Any request that is neither an incident or an RFC (Request for Change).
- Used for simple operations on the state of a configuration item.
 - **For example:** Stopping or Starting a service / Reset a password.
- No impact on Charge Back.

Request for Change ([RFC](#))

- A request for the creation of a new architecture or modification to an existing architecture.
 - **For example:** Requesting new configuration items and removing and amending service elements (Publish Map Services or restart a Map Service).
- Can impact Charge Back.

How to make an RFC / RFS request

1. Make an [RFC](#) or [RFS](#) in [JASSPR](#).
 - a. Complete the fields in the form.
 - b. Include as much useful information as possible.
 - c. Submit the form.
 - d. Monitor the request status on the [My requests list](#).
 2. For information and examples on how to use [JASSPR](#) - consult the [JASSPR User Guide](#).
-

What is an RFI?

Request for Information (RFI)

Use an [RFI form](#):

- In cases when help is required to define exactly how an envisioned architecture should be requested.
- When you are unable to find the required service.



- DIGIT works to ensure this service is as extensive as possible and covers customers requirements and expectations.
- If a customer requires an additional option or configuration to be set up, DIGIT will study the RFI request for its feasibility, before delivering an answer.

How to make an RFI request

1. Open [JASSPR](#).
2. Select **Information Request** from the left panel.
3. Complete the fields in the RFI form.
4. Provide as much useful information as possible in the Description field.
5. Submit the request.
6. Monitor the request status through [My Requests list](#).

How to find more information

Watch one of the following JASSPR [tutorial videos](#) to learn more.

SAP Data Services: How to request the creation or modification of an instance



Homepage

1. Make an [RFC](#) request in [JASSPR](#) (Create instance, Add Node, Remove Node)
 - a. Complete the fields in the form
 - b. Include as much useful information as possible.
 - c. Submit the form.
 - d. Monitor the request status on the [My requests list](#).
2. For information and examples how to use [JASSPR](#) - consult the [JASSPR User Guide](#)

Related Links

[What requests can you make?](#)

Refer to the [Tutorial videos](#) in JASSPR to learn how to create an RFC or other requests.

SAP Data Services: Learn about your service

- [SAP Data Services: Backup & restore](#)
- [SAP Data Services: Upgrading the service](#)
- [SAP Data Services: Versions & Patches](#)
- [SAP Data Services: Licenses](#)
- [SAP Data Services: How to decommission the service](#)

SAP Data Services: Backup & restore



Homepage

Page Topics

- [Important Notice: Backup and Restore](#)
- [What is the frequency of backups?](#)
- [What is the back-up retention period?](#)
- [What is backed-up?](#)
- [Who manages the backup?](#)
- [Why is a restore performed?](#)
- [What can be restored?](#)
- [What is the restoration period?](#)

Related Links

- [Upgrading the service](#)
- [Versions & Patches](#)
- [Licenses](#)
- [How to decommission the service](#)



Important Notice: Backup and Restore

Performing a full backup and restore of the system can be **complex and error-prone**. We strongly recommend considering **mitigation strategies** to reduce risk.

As a best practice, each developer should **regularly export their repositories in ATL format** to their **local machine** to ensure individual backups are available if needed.

What is the frequency of backups?

Daily backups.

What is the back-up retention period?

Data retention of one month.

What is backed-up?

File systems (NAS & SAN)

Who manages the backup?

DIGIT

Why is a restore performed?

- To restore lost or damaged data.
- Ensures data backup is consistent with the state of the data at a specific point in time.

What can be restored?

Any file associated with an instance.

What is the restoration period?

Any day within the last 4 weeks.

SAP Data Services: Upgrading the service



Homepage

Page Topics

- [What is the frequency of upgrades?](#)
- [Why is an upgrade performed?](#)
- [How does DIGIT perform an upgrade?](#)

Related Links

- [Backup & restore](#)
- [Versions & Patches](#)
- [Licenses](#)
- [How to decommission the service](#)

What is the frequency of upgrades?

- DIGIT organises and communicates planned upgrades in advance to the customer.
- DIGIT continuously evaluate new upgrades. Once a new release is confirmed, DIGIT will release the new upgrade as soon as possible.

Why is an upgrade performed?

- Fixes a problem in the current version.
- Reduces security vulnerability.
- The current version is no longer supported.
- A request from the customer is received to upgrade.



SAP Data Services architecture rely on network components such as RPM and LoadBalancers that can be subject to migration or upgrade.

How does DIGIT perform an upgrade?

PHASE	DESCRIPTION
Plan and Prepare	DIGIT plans the upgrade and ensures the defined supported standards are not affected by compatibility issues.  Refer to the Prerequisites for the supported standard details.
DIGIT Validation	All non-production environments are upgraded to the target version by DIGIT.
Customer Validation	The customer validates the environments.
Production Environment	The PRD environment is upgraded (Performed by DIGIT)

SAP Data Services: Versions & Patches



Homepage

Page Topics

- [What versions of SAP Data Services are supported by DIGIT?](#)
- [What patches are installed?](#)
- [Is downtime required for patching?](#)

Related Links

- [Backup & restore](#)
- [Upgrading the service](#)
- [Licenses](#)
- [How to decommission the service](#)

What versions of SAP Data Services are supported by DIGIT?

- Check the [Services Roadmap Dashboard](#) to view the lifecycle of the service.
Select:
 1. Service Roadmap Dashboard
 2. General Roadmap
 3. General Roadmap
- View information about:
 - End of support
 - Deprecated dates
 - Working duration
 - Availability of new services



Refer to the Services Roadmap Dashboard [User Guide](#).

The upgrade must be on the official list supported by DIGIT.

What patches are installed?

Installation of security or technical patches.

Is downtime required for patching?

- Outage might occur while OS patching and a reboot may be required.
- Automatic scripting will launch all SAP Data Services components after a reboot.

SAP Data Services: Licenses



Homepage

Page Topics

- [What does the license cover?](#)
- [What are the license restrictions?](#)
- [How to find more information](#)

Related Links

- [Backup & restore](#)
- [Upgrading the service](#)
- [Versions & Patches](#)
- [How to decommission the service](#)

What does the license cover?

- IPS (Information Platform Services)
- Data Services and Web application servers
- SAP Data Services

What are the license restrictions?

- The enterprise license agreement is for the European Commission.
- This is a “Single Metric License” based on the total number of employees in the commission. There are no restrictions concerning your number of users and servers.
- The license cannot be used by entities that are legally independent of the European Commission. If you are in the latter case, you must bring your own license.

How to find more information

For any licensing questions, contact [DIGIT ISHS BO BI](#).

SAP Data Services: How to decommission the service



Homepage

Page Topics

- [What is decommissioned?](#)
- [How to request a decommission](#)

Related Links

- [Backup & restore](#)
- [Upgrading the service](#)
- [Versions & Patches](#)
- [Licenses](#)

What is decommissioned?

- SAP Data Services Instance
- DIGIT provides a copy of the data on request before the decommissioning process.

How to request a decommission

1. Make an [RFC](#) in [JASSPR](#) (Decommission SAS Instance)
 - a. Complete the fields in the form.
 - b. Submit the request.
 - c. Monitor the request status on the [My requests list](#).
2. For information and examples how to use [JASSPR](#) - consult the [JASSPR User Guide](#)

SAP Data Services: Authentication



Homepage

Page Topics

- [Who is responsible for performing authentication?](#)
- [What is the authentication and rights management?](#)
- [What is the Purpose of Security Plug-ins?](#)
- [What authentication methods are supported?](#)
- [How is authentication achieved?](#)
- [Can the Data Services platform be customized?](#)
- [What security does DIGIT provide?](#)

Related Links

[SAP Data Services: Tools](#)

Who is responsible for performing authentication?

DIGIT performs and supports the required configuration steps regarding the authentication.

What is the authentication and rights management?

- Authentication is the process of verifying the identity of a user who attempts to access the system.
- Rights management is the process of verifying that the user has been granted sufficient rights to perform the requested action upon the specified object.

What is the Purpose of Security Plug-ins?

Security plug-ins allow the Business Intelligence (BI) platform to **extend and customize authentication mechanisms**, enabling integration with third-party identity systems.

They facilitate user and group management by allowing you to **map external accounts** into the BI platform environment.

With security plug-ins, you can:

- Map third-party **user accounts or groups** to existing BI platform accounts or groups
- Automatically **create new Enterprise accounts or groups** corresponding to external directory entries
- Streamline **account provisioning and access control** across integrated systems

What authentication methods are supported?

DIGIT **strongly recommends** that you use [EU Login](#) for authentication purposes.

The current release supports the following authentication methods:

- [EU Login](#)
- [LDAP](#)
- Enterprise

How is authentication achieved?

LDAP is used as the source for authentication. (This is the case for EU Login and LDAP. The enterprise option is linked either automatically or manually)

Can the Data Services platform be customized?

The Data Services platform is fully customizable and the authentication and processes may vary from system to system.

What security does DIGIT provide?

- DIGIT aims to keep an all-round secure infrastructure. Application vulnerabilities are however the customer's responsibility.
- It can be necessary for systems to undergo Vulnerability Testing and accordingly DIGIT teams, with the support of the [DIGIT TEST CENTRE](#) team can advise on how best to fix potential issues.
- The customer can also make a request to the [DIGIT TEST CENTRE](#) to initiate Security Testing.

SAP Data Services: Tools

- [SAP Data Services: Designer tool](#)
- [SAP Data Services: Extract-Transform-Load \(ETL\) tool](#)
- [SAP Data Services: Management Console](#)

SAP Data Services: Designer tool



Homepage

Page Topics

- [What is the purpose of the Designer?](#)
- [How to Use the Designer](#)
 - [Metadata](#)

Related Links

[Management Console](#)

[Extract-Transform-Load \(ETL\) tool](#)

What is the purpose of the Designer?

- The Designer is a development tool with an easy-to-use graphical user interface.
- It enables developers to define data management applications that consist of data mappings, transformations, and control logic.
- Use the Designer to create applications containing work flows (job execution definitions) and data flows (data transformation definitions).

You can install the Designer virtual client via the **EC Store**.

How to Use the Designer

The **SAP Data Services Designer** allows users to create and configure objects by **dragging, dropping, and editing icons** within flow diagrams, table layouts, and nested workspace pages.

It is now possible to **execute jobs directly from the Designer**, without needing to access the SAP DS Management Console.

To enable this functionality, the **IP address of the SAP BO DS Server** must be added to the firewall object **DIGIT_SAP_BI_T2-CALLBACK**, which authorizes client/server communication for the Designer tool.

This configuration requires submitting a request via **AlgoSec** to update the callback rule with the new IP address of the SAP BO DS Server.

Metadata

- The objects in the Designer represent metadata.
- The Designer interface enables you to manage metadata stored in a repository.
- From the Designer, you can also trigger the Job Server to run your jobs for initial application testing.

SAP Data Services: Extract-Transform-Load (ETL) tool



Homepage

Page Topics

- [What is the ETL tool?](#)
- [What is the role of the ETL tool?](#)
- [How does the ETL tool work?](#)

Related Links

- [Management Console](#)
- [Designer tool](#)

What is the ETL tool?

- **Extract-Transform-Load (ETL)** is an essential function of any data warehouse (DWH).
- In a majority of DWH projects, the usage of an ETL solution is the most cost- and time-effective approach to setup and maintain the data feeding process.

What is the role of the ETL tool?

The following components will likely reside on separate server hosts and possibly different platforms:

- Source databases
- Staging area
- DWH database/appliance

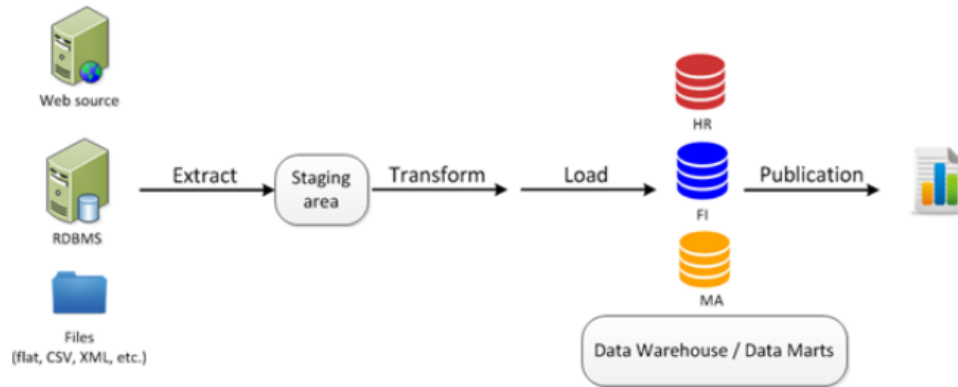
The role of an ETL application is to link the above components together in order to move data from one system to another.

How does the ETL tool work?

The nature of the ETL system is straightforward:

1. Extract data from operational databases/systems.
2. Transform data according to the requirements of your data warehouse so that the different pieces of data can be used together.
3. Apply data quality transformation methods in order to cleanse data and ensure that it is reliable before it gets loaded into a data warehouse.

4. Load conformed data into a data warehouse so that end users can access it via reporting tools, using client applications directly.



SAP Data Services: Management Console



Homepage

Page Topics

- [What is the Purpose of the Management Console?](#)

Related Links

- [Designer tool](#)
- [Extract-Transform-Load \(ETL\) tool](#)

What is the Purpose of the Management Console?

The **SAP Data Services Management Console** provides a **browser-based interface** for administrators to manage and monitor Data Services resources.

Key administrative functions include:

- Scheduling, monitoring, and executing **batch jobs**
- Configuring, starting, and stopping **real-time services**
- Managing the configuration of the **Job Server**, **Access Server**, and **repository usage**
- Configuring and managing **adapters**
- Managing **user access and permissions**
- Publishing **batch and real-time services** as **Web services**

SAP Data Services: Repository & Staging Areas

- [SAP Data Services: SAP Data Services repository](#)
- [SAP Data Services: Staging & Sharing Areas](#)

SAP Data Services: SAP Data Services repository



Homepage

Page Topics

- [What is the SAP Data Services Repository?](#)
- [What are the types of repository?](#)
- [Where are the repositories stored?](#)
- [What is the purpose of a local repository?](#)
- [What is the associated environment?](#)
- [What is the purpose of the central repository?](#)
 - [Job Execution](#)
- [What is the associated environment?](#)
- [What is the purpose of the Profiler repository?](#)
- [What is the associated environment?](#)

Related Links

[Staging & Sharing Areas](#)

What is the SAP Data Services Repository?

The **SAP Data Services repository** is a collection of **database tables** that store:

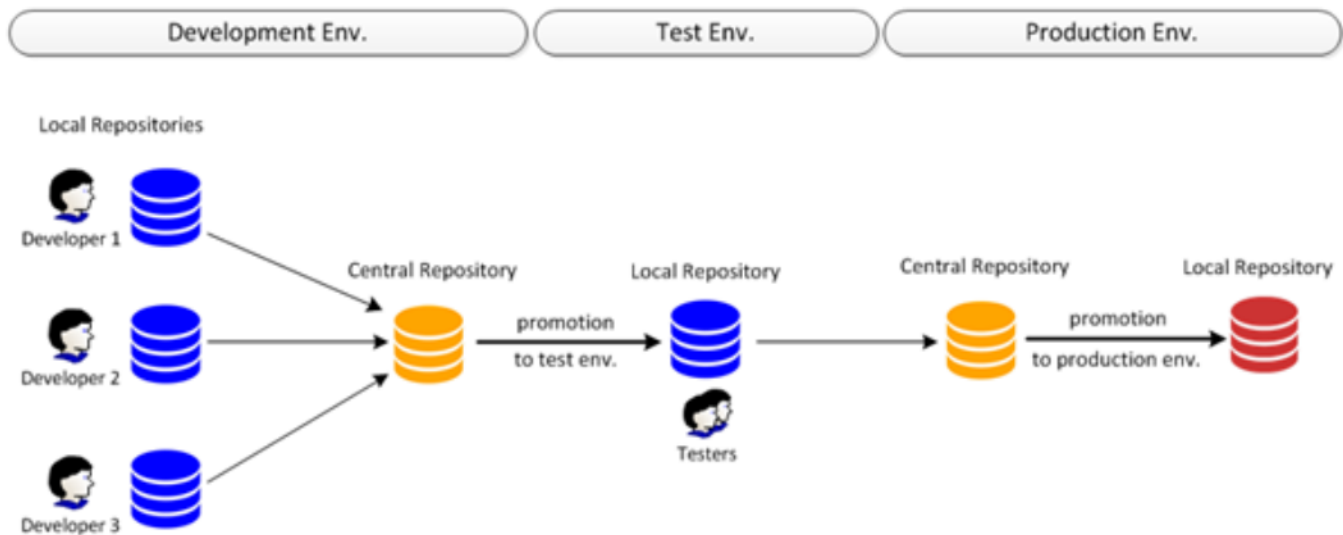
- User-created and predefined **system objects**
- **Source and target metadata**
- **Transformation rules** and job definitions

Repositories are set up on an **open client/server platform**, enabling integration and **metadata sharing** with other enterprise tools across the organization.

What are the types of repository?

There are three types of repositories:

- Local
- Central
- Profiler



Where are the repositories stored?

- Each repository must be stored on an existing RDBMS and registered in the Central Management Console (CMC).
- Each repository is associated with one or more Job Servers, which run the jobs you create.

What is the purpose of a local repository?

- A local repository is the workspace of an ETL (Extract-Transform-Load) developer where the unit testing of ETL jobs is developed and performed.
- The term "local" is specific to SAP Data Services terminology.
- Only one version of each ETL object is stored in a local repository.

What is the associated environment?

- Each local repository is associated with one ETL developer for a Development environment.
- A single common local repository is delivered for the Test and Production environments.

What is the purpose of the central repository?

- A central repository is used to share objects, and to keep different versions of each object.
- The ETL designers can connect to the central repository to add and get versions of the objects.

Job Execution

- Jobs are not possible to be executed from a central repository. Import them first to a local repository associated to the job server.
- Objects in the central repository cannot be changed directly, they are versioned with check/in-check/out.

What is the associated environment?

DIGIT delivers only two central repositories for each DG/IS:

- The first is used in the promotion of objects from the Development to Test environment.
- The second is used to promote the objects from the Test to the Production environment.

What is the purpose of the Profiler repository?

The profiler repository is used to profile source data. It collects statistical information about selected tables and columns.

- Counts
- Value ranges
- Patterns



The profiler requires a separate repository to store profiler information.

What is the associated environment?

DIGIT offers one profiler repository for each of the Development, Test and Production environments.

SAP Data Services: Staging & Sharing Areas



Homepage

Page Topics

- [What is a Staging Area?](#)
- [What are the benefits of using a Staging Area?](#)
- [What is a Sharing Area?](#)
- [How can developers access flat files?](#)

Related Links

[SAP Data Services repository](#)

What is a Staging Area?

An intermediate and temporary storage area used for data processing during the extract, transform and load (ETL) process

What are the benefits of using a Staging Area?

The data staging area sits between the data source(s) and the data target(s), which are often data warehouses, [data marts](#), or other data repositories.

A Staging Area is beneficial for:

- Delta-loading
- Transformation needs
- De-coupling
- Debugging
- Failure recovery

What is a Sharing Area?

A Sharing Area is available for flat files to be shared and used as a source or target datasource.

How can developers access flat files?

- SAP Data Service developers can access flat files, available using sftp (Dev) or ssh (Test and Prod).
- Currently SAP Data Services supports excel workbooks, xml schemas, and flat files

SAP Data Services: Monitoring



Homepage

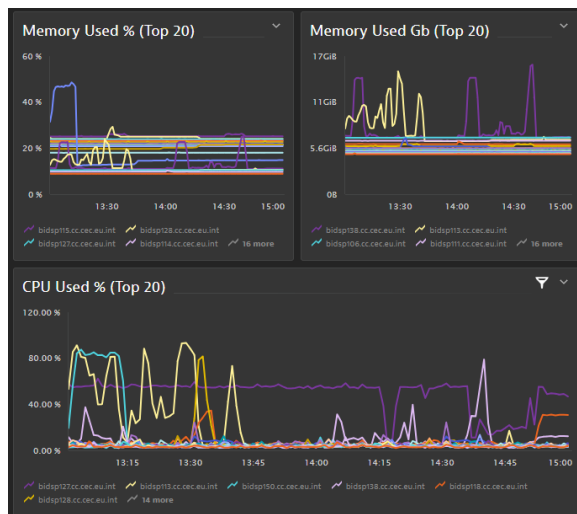
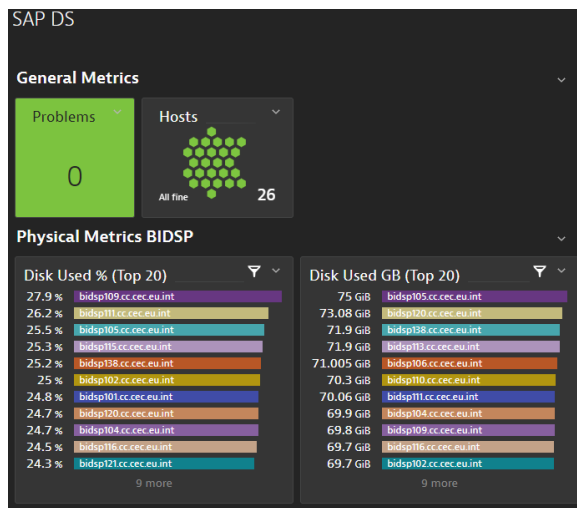
What type of monitoring is available?

- The attached job (ATL file) is designed to confirm the availability of the SAP Data Services instance.
- This job will be scheduled on the server and generates a log file that is monitored by DIGIT ISHS MONITORING team every 10 minutes.
- It's available in Dynatrace the dashboard to monitor the CPU, Disk and Memory of each SAP DS node: [Dynatrace Monitoring - SAP DS Dashboard](#)

Related Links

[Support & Useful Links](#)

[FAQ](#)



[BODS_monitoring.atl](#)

SAP Data Services: FAQ & Supporting Links

- [SAP Data Services: FAQ](#)
- [SAP Data Services: Support & Useful Links](#)

SAP Data Services: FAQ

- [SAP Data Services: General FAQ](#)
- [SAP Data Services: General Recommendations](#)
- [SAP Data Services: How to connect EUSend with SAP Data Services](#)
- [SAP Data Services: How to connect SUMMA DWH with SAP Data Services](#)
- [SAP Data Services: How to use PrivX with SAP Data Services](#)

SAP Data Services: General FAQ



Homepage

Page Topics

- [Check the Change Log to see what's new and track past changes](#)
- [What are the Service components?](#)
- [What is the prerequisite knowledge?](#)
- [What is the deployment architecture?](#)
- [What does the architecture comprise?](#)
- [What comprises the Production architecture?](#)
- [How is High Availability supported?](#)
- [Where can I find chargeback information?](#)

Related Links

- [General Recommendations](#)
- [How to connect EUSend with SAP Data Services](#)
- [How to connect SUMMA DWH with SAP Data Services](#)
- [How to use PrivX with SAP Data Services](#)

NEW

Check the [Change Log](#) to see what's new and track past changes

What are the Service components?

The main components of the service are:

- Information Platform Services (IPS)
- Data Services & Web Application servers
- SAP Data Services & Service Pack



Refer to the [Services Roadmap Dashboard](#) for component versions.

What is the prerequisite knowledge?

It is essential that customers making requests have a sound understanding of the SAP Business Intelligence Platform technology and the technology it depends on.

- SAP BusinessObjects client tools;
- Basic knowledge of SQL

What is the deployment architecture?

- One stand-alone virtual machine for Development and Test environments.
 - Two dedicated Virtual Machines (VMs) for Production environments.
-

What does the architecture comprise?

Each SAP Data Service instance:

- Is deployed on one (Stand-alone) Virtual Machine (VM) or a cluster of VMs (Clustered).
- Runs on the Red Hat Enterprise Linux operating system.

A SAP Data Service instance is made of three main components:

- IPS (Information Platform Services)
 - Data Services
 - Web Application servers (The default web application server is Tomcat)
-

What comprises the Production architecture?

- Each SAP Data Services instance is deployed using two dedicated Virtual Machines (VMs)
 - The sizing of the VM is based on the ETL (Extract-Transform-Load) expected workload and DIGIT license compliance.
 - Environments are supported 24/7
-

How is High Availability supported?

- Runs on a pair of Virtual Machines (VMs) split across datacentres.
 - If a server or a datacentre becomes unavailable, high availability is maintained by running on one leg of the architecture (Using a [Load Balancer](#) and [Virtual Machine \(VM\) pairing](#)).
-

Where can I find chargeback information?

Refer to the [DIGIT Infrastructure Services price list](#).

SAP Data Services: General Recommendations



Homepage

Page Topics	Related Links
SAP DS Recommendations Governance Plan Review – SAP Data Services	General FAQ How to connect EUSend with SAP Data Services How to connect SUMMA DWH with SAP Data Services How to use PrivX with SAP Data Services

SAP DS Recommendations



SAP Data Services (SAP DS) can be resource-intensive, utilizing significant CPU, I/O (Disk/Network), and Memory. These resources are critical and must align with operational requirements.

Given the fixed infrastructure for our servers—comprising 4 CPUs and 64 GB of RAM in the production environment—it's essential to adhere to the following key recommendations when developing ETL processes:

Governance Plan Review – SAP Data Services

A thorough review of the governance plan for **SAP Data Services (SAP DS)** should cover the following key elements to ensure operational efficiency, resource optimization, and alignment with best practices:

1. Repository and Developer Management

- Evaluate the **number of repositories** and **active developers** to ensure proper resource allocation and avoid redundancy.

2. Usage Monitoring and Resource Adaptation

- Monitor **usage statistics** regularly.
- Adjust **process execution plans** based on system capacity and resource availability.

3. Job Execution Strategy

- Review **job requirements** and define clear **execution priorities** to optimize job scheduling and runtime performance.

4. Environment Execution Planning

- Develop and maintain a well-documented **execution plan** for all environments (development, testing, production) to support consistency and traceability.

5. Development Content Governance

- Ensure that **only active jobs** are enabled in the **local repository**.
- Store all development work in the **Central Repository** to maintain version control and collaboration.
- Archive inactive or deprecated developments using **ATL exports**.

6. Resource Consumption Review

- Regularly assess job performance in **pageable** and **in-memory** modes.
- Identify and resolve excessive memory or disk consumption issues.

7. Daily Maintenance

- Implement automated scripts for **daily cleanup** of **PCache** and other temporary ETL artifacts to maintain system hygiene.

8. Data Dump Control

- Prevent uncontrolled **data dumps** from sources to staging areas.
- Enforce **delta logic** to manage data volume and storage efficiency.

9. Best Practices Alignment

- Review and document current practices.
- Ensure alignment across the team for any **developments not yet implemented** or deviating from standards.

By adhering to these governance practices, the team can ensure a **stable, scalable, and maintainable ETL environment** within SAP Data Services.

SAP Data Services: How to connect EUSend with SAP Data Services



Homepage

Page Topics

- [What are the EU Send sFTP Integration Instructions?](#)
 - [Preparing EU Send](#)
 - [Key Generation](#)
- [Designer - File Location](#)

Related Links

- [General FAQ](#)
- [General Recommendations](#)
- [How to connect SUMMA DWH with SAP Data Services](#)
- [How to use PrivX with SAP Data Services](#)



EU Send is offered to EU Institutions and Agencies for **exchanging electronic data and/or documents in a secure way** with the following:

- EU Institutions;
- EU Agencies;
- Any private or public entity in Member States and third countries.
- **EU Send sFTP** enables the secure transfer of large files as an integral part of the larger [EU Send](#) service.
- The SAP Data Services service is concerned with the integration of **EU Send sFTP** as a new data source for our SAP BO Data Services platforms.

Domain Administration

- **EU Send sFTP** clients are assigned domains for which they have a limited responsibility via a Domain Administrator account.
- Domain administrators are given the responsibility of creating and managing users and their assigned folders.
- A Web Interface enables Domain administrators and users to operate sFTP for uploading and downloading files.

Find further detailed information in the user guide section at [EU Send Documentation - sFTP](#).

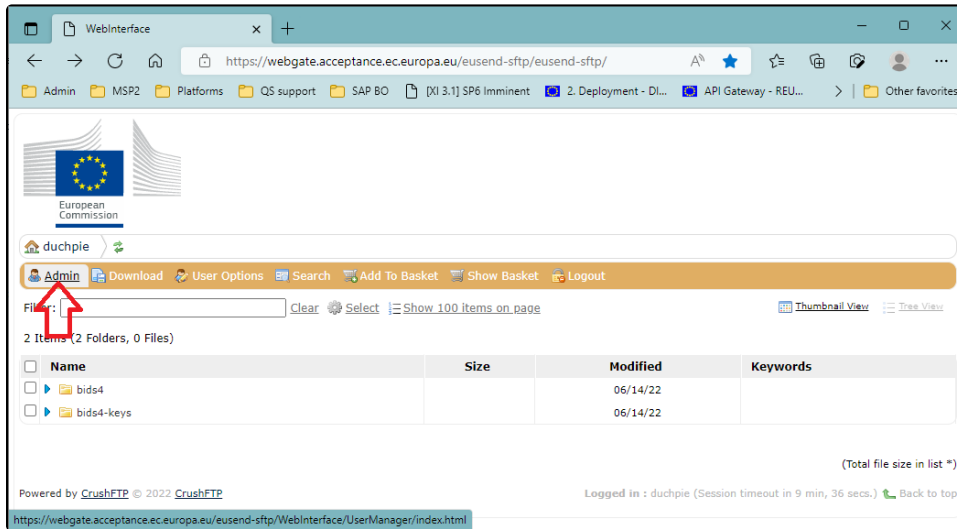
What are the EU Send sFTP Integration Instructions?

Preparing EU Send

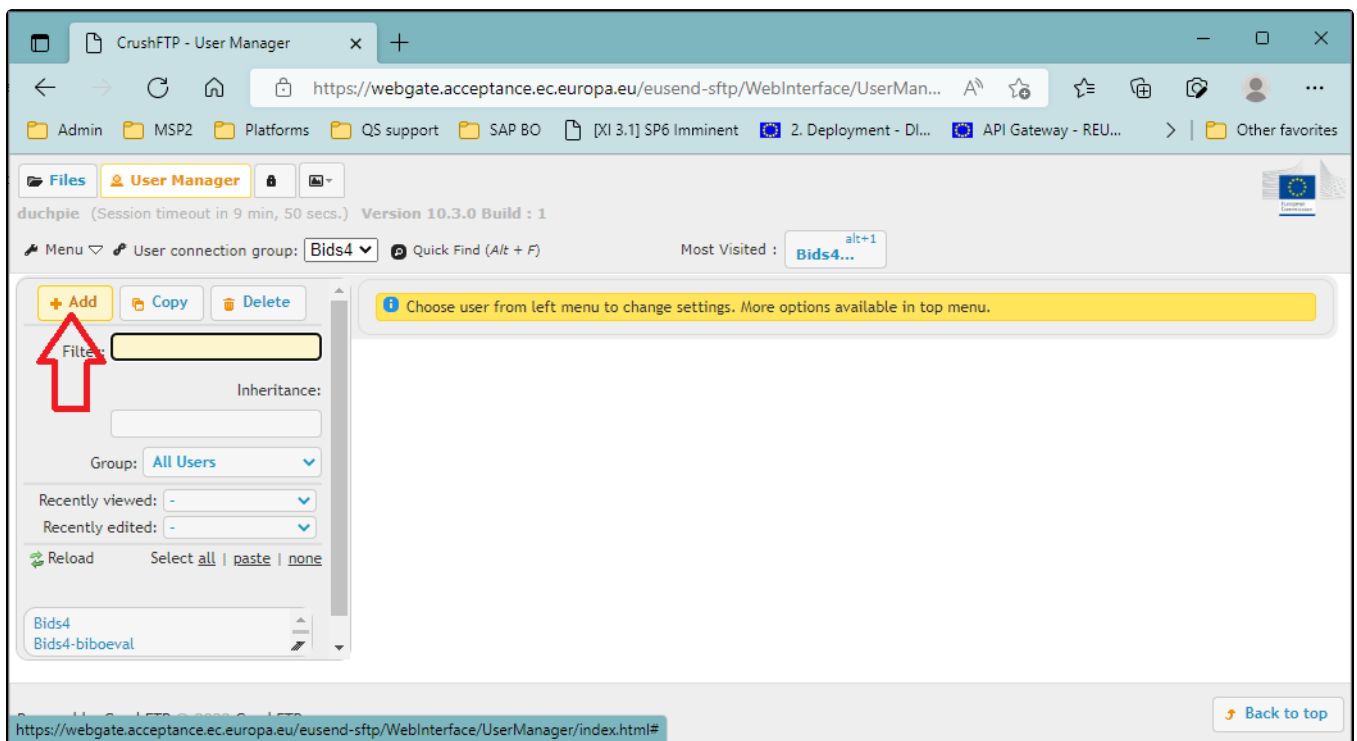
To interact with EU Send sFTP, a user and folder are required. These are created in the [WebInterface](#) of EU Send.

1. Login using the domain admin account provided by the EU Send service team.

2. Click on the 'Admin' button.

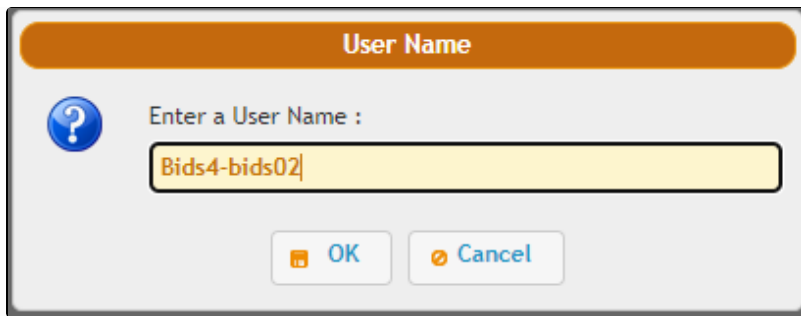


3. In the User Management screen click on 'Add'.



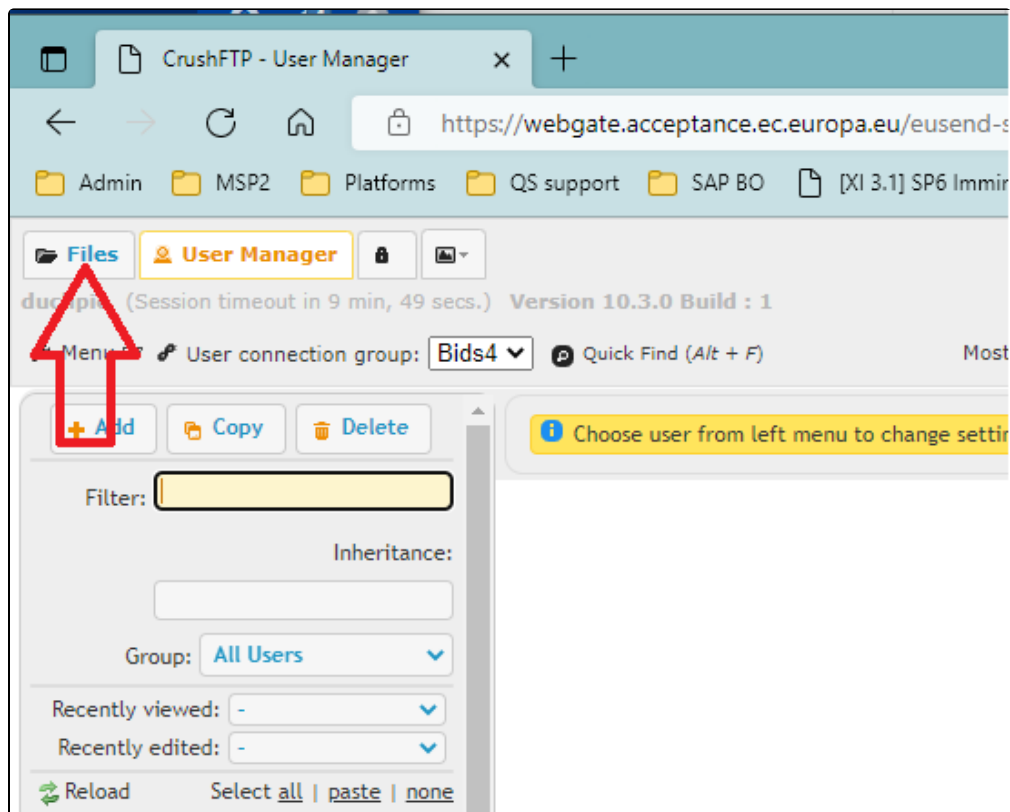
i It is recommended to create a user in order to follow the naming convention of Domain name followed by a dash and then a distinct user name.

As an example opposite, for domain bids4, the user name biboeval (technical user for evaluation purposes) gives us *Bids4-biboeval*.



A dialog box titled "User Name" with a question mark icon. It contains the text "Enter a User Name :" followed by a text input field containing "Bids4-bids02". At the bottom are "OK" and "Cancel" buttons.

4. Return to the main screen by clicking on the 'File' button.

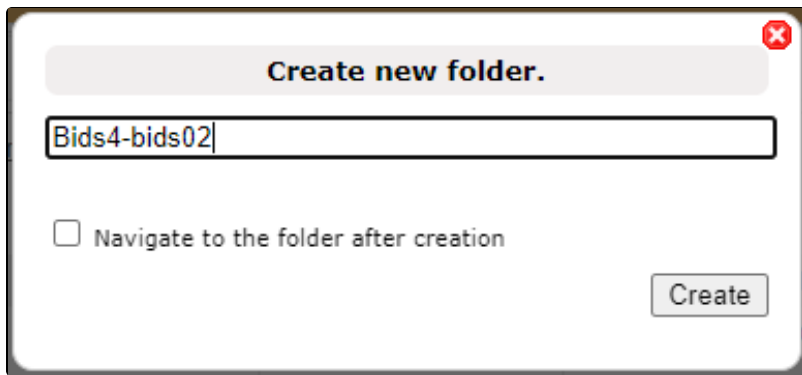


5. From the main screen, click on the 'Domain' folder.

6. Click on the 'Create Folder' button.

7. Create a new sub-folder in the Domain folder and name it the same as the user you have just created

8. Click the 'Create' button.

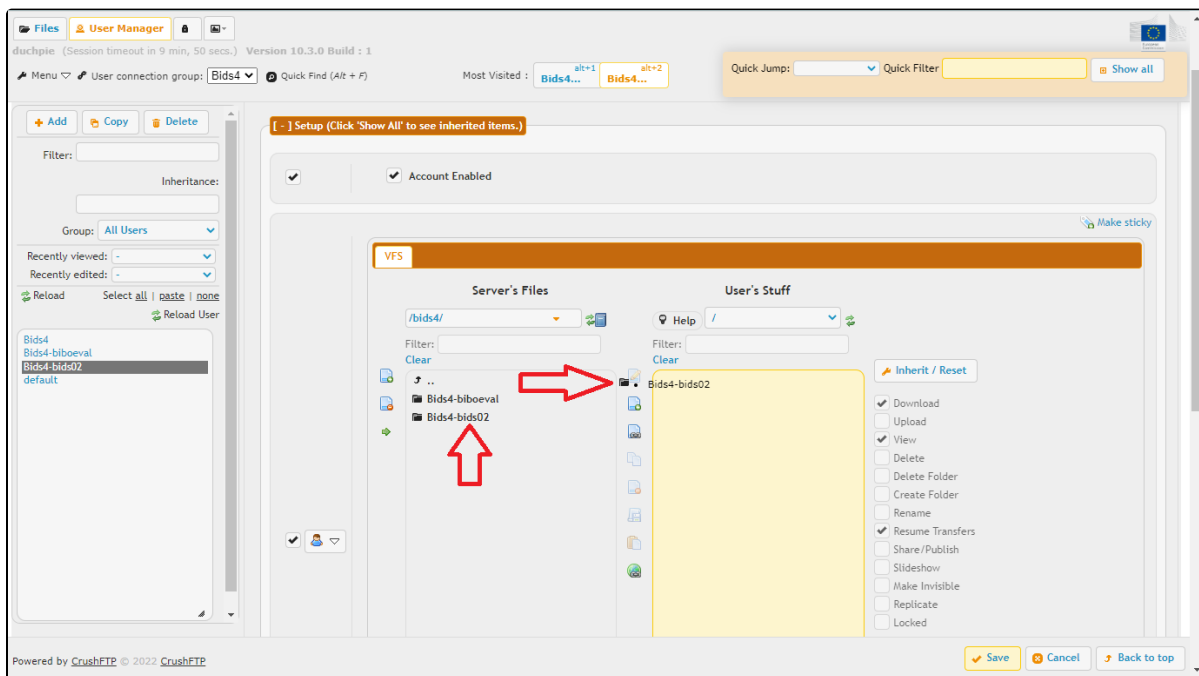


9. Return to the main screen and click the 'Admin' button.

10. Select you user in the listed user portion of the screen.

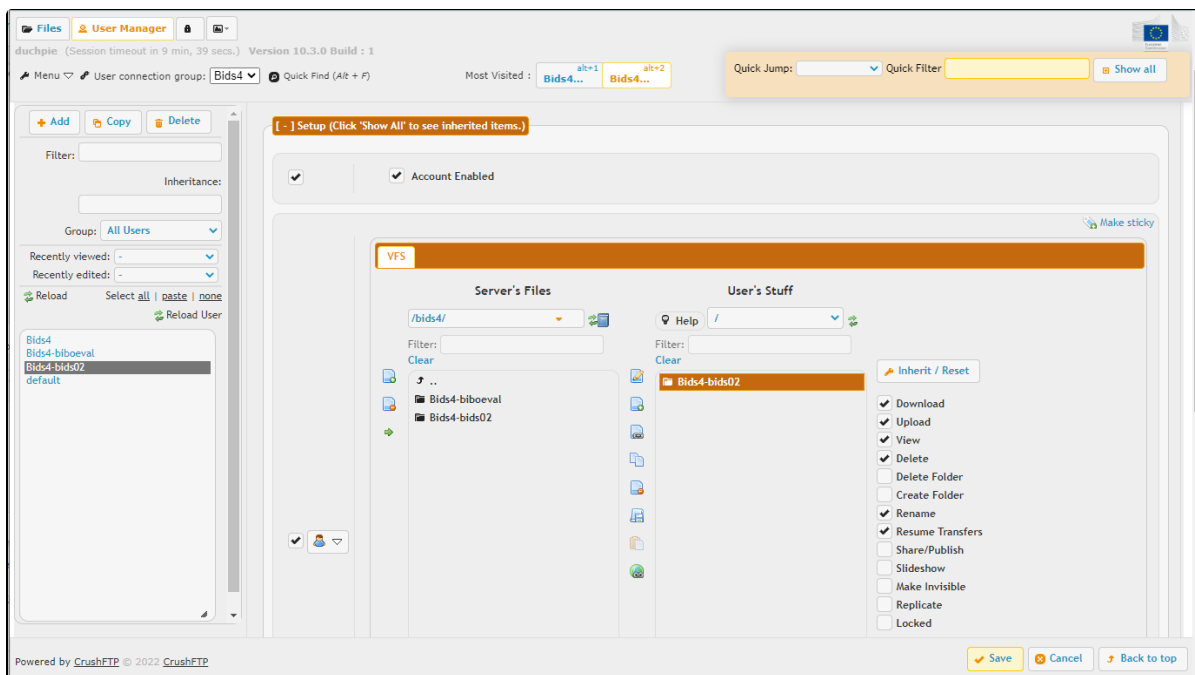
11. Navigate to the newly created sub-folder under the 'Server's Files' portion of the screen.

12. Select the sub-folder under the 'Server's Files' portion of the screen and drag it over to the right hand side under 'User's Stuff'.



The user rights to Download, Upload, View, Delete, Rename and Resume Transfer can be assigned.

13. Click the 'Save' button.



Key Generation

1. Using the DOS Command prompt, generate a ECDSA type key pair for the user.
2. Type the following at the DOS prompt:
3. Enter the EU Send user name for input to the 'Enter file in which to save the key:

(C:\Users\<user name>\.ssh/id_ecdsaz):

```
C:\Users\duchpie>ssh-keygen -t ecdsa -b 256
Generating public/private ecdsa key pair.
Enter file in which to save the key (C:\Users\duchpie\.ssh/id_ecdsa): Bids4-bids02
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in Bids4-bids02.
Your public key has been saved in Bids4-bids02.pub.
The key fingerprint is:
SHA256:N0zI0yr8iGYTSifh7sIMlGZeDVgPzPm+vKKfNpp9qd8 net1\duchpie@L193660994
The key's randomart image is:
+---[ECDSA 256]---+
|  oo.                |
| o+o . o            |
| o ... + o          |
| . o +. =           |
| O +.+ S +         |
| B = o.+ . .       |
| = =..O.            |
| = *+O=.            |
| O==*O.E            |
+---[SHA256]-----+
```

4. Carefully note the key passphrase for later use.
5. Use Windows Explorer to navigate to the user directory and locate the newly generated Public key (.pub) and the Private key (no file type).

6. Copy these to keys to a storage location.

7. Log onto your staging area's file system using a terminal window or WinSCP.

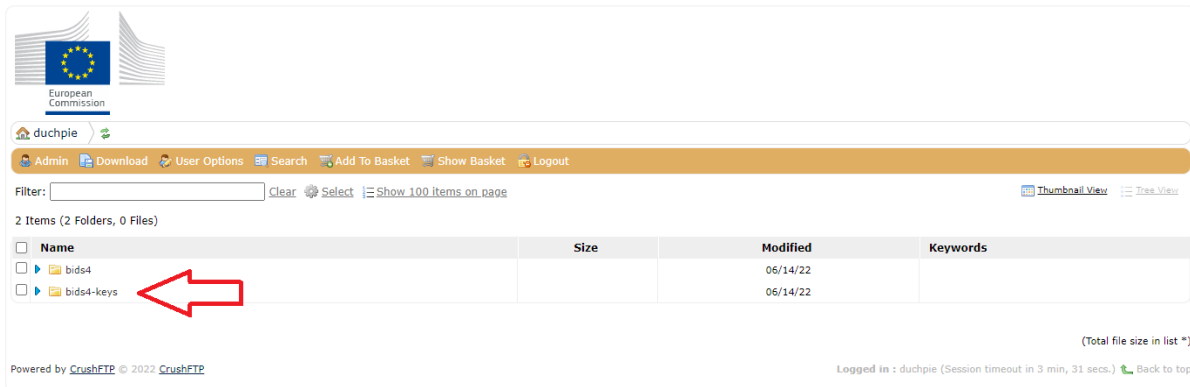
8. Create a keys folder to store your keys:

```
/ec/<environment>/server/home/bods/stage/<UserAccount>/keys/
```

9. Place a copy of both the private and public keys in the folder just created to make them available to developers for later File Location object referencing.

10. Log into [WebInterface](#) of EU Send as domain admin.

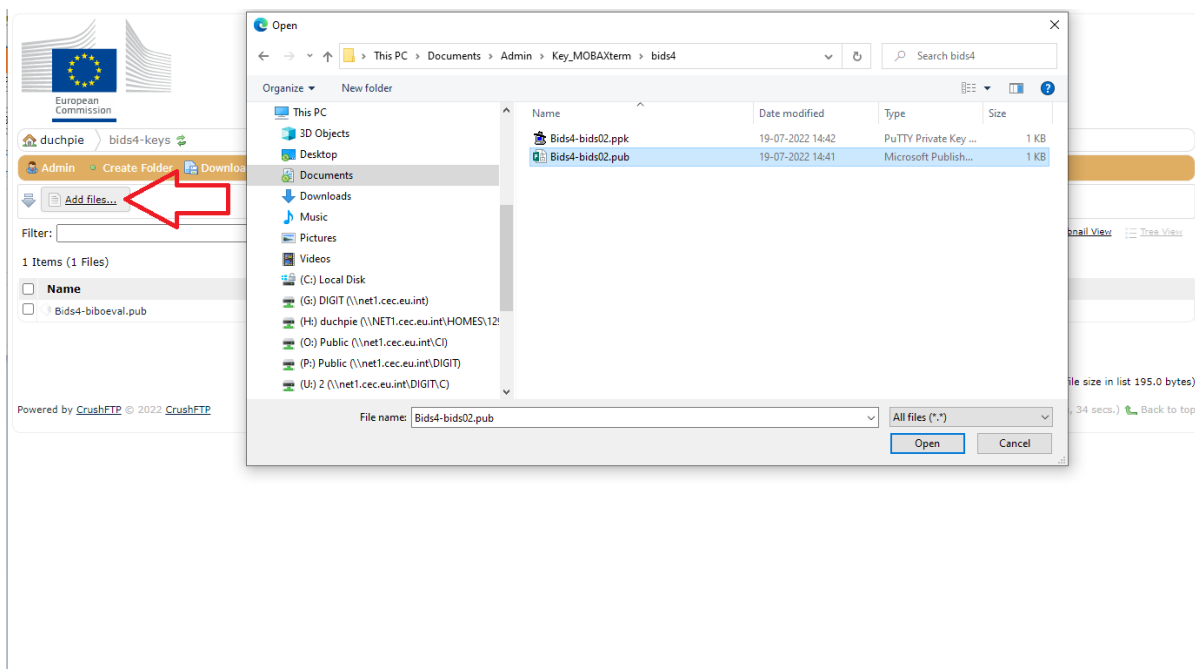
11. From the main screen, click on the folder "domain"-keys (ie, bids4-keys in image opposite)



This will open the keys folder.

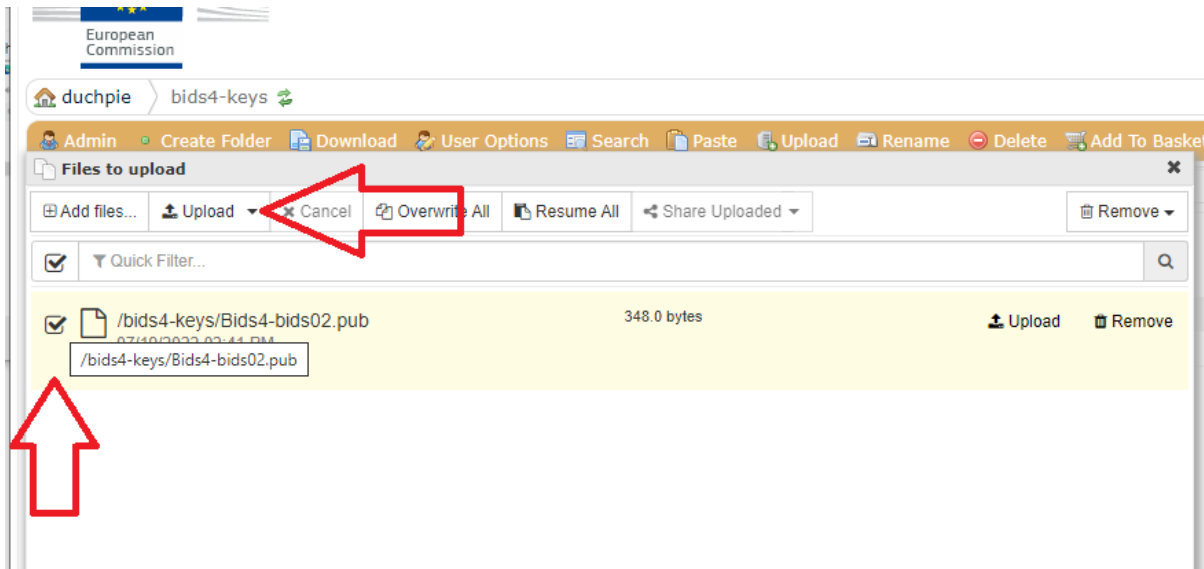
12. Click on the 'Add files' button.

13. Select the public key (.pub).

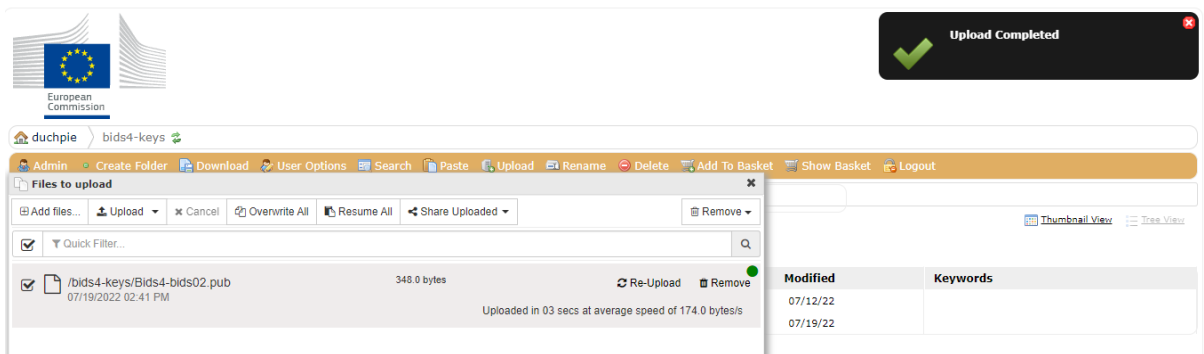


14. Click the checkbox to select the public key file.

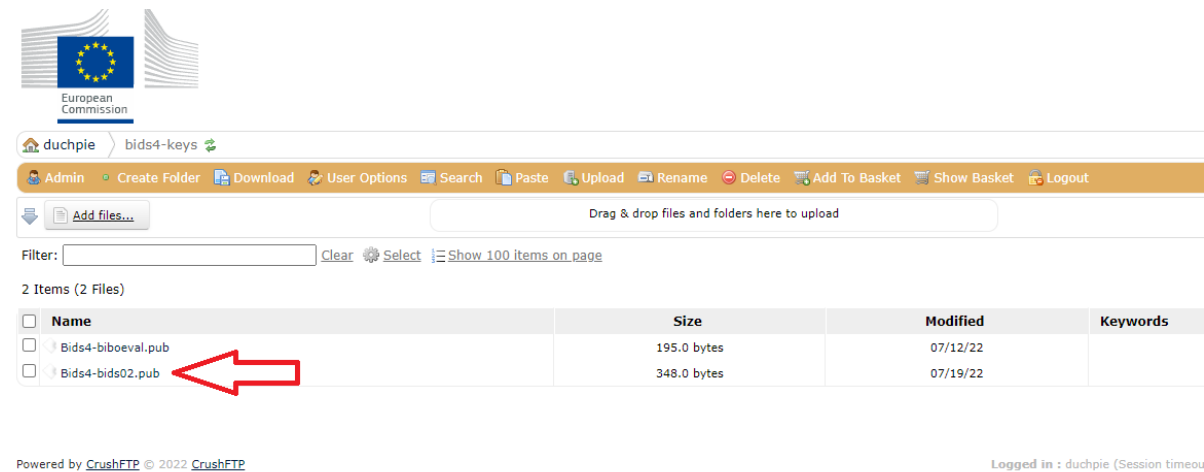
15. Click the 'Upload' button.



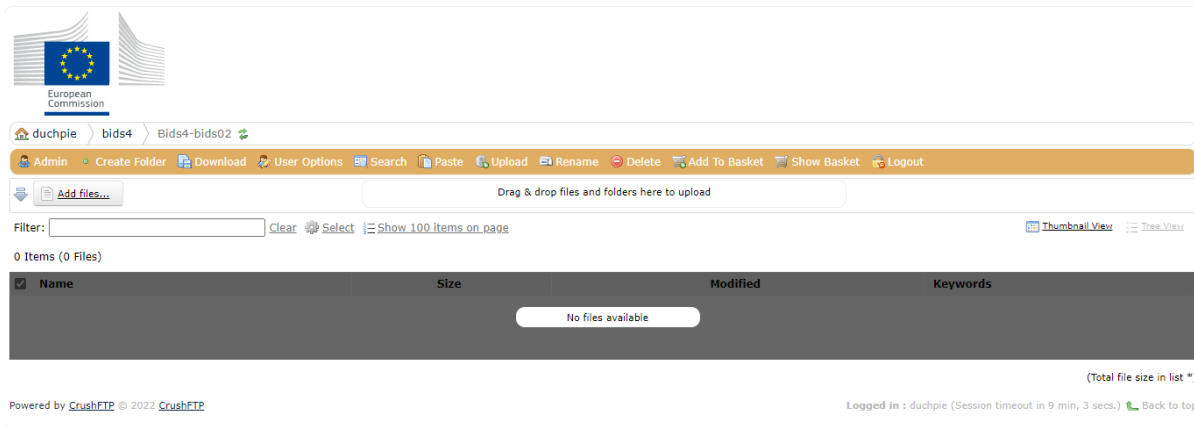
An 'Upload completed' message is displayed upon successful upload.



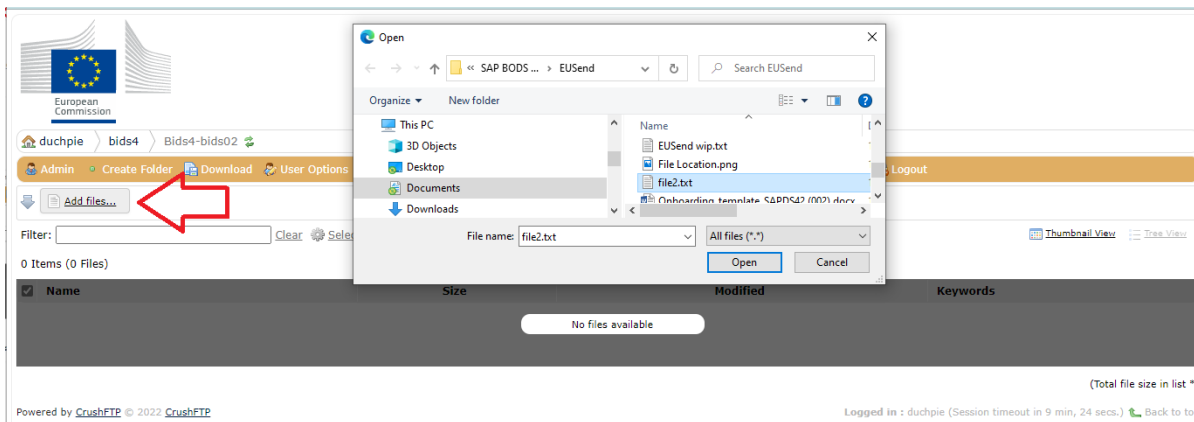
The public key will now be displayed in the domain keys sub-folder.



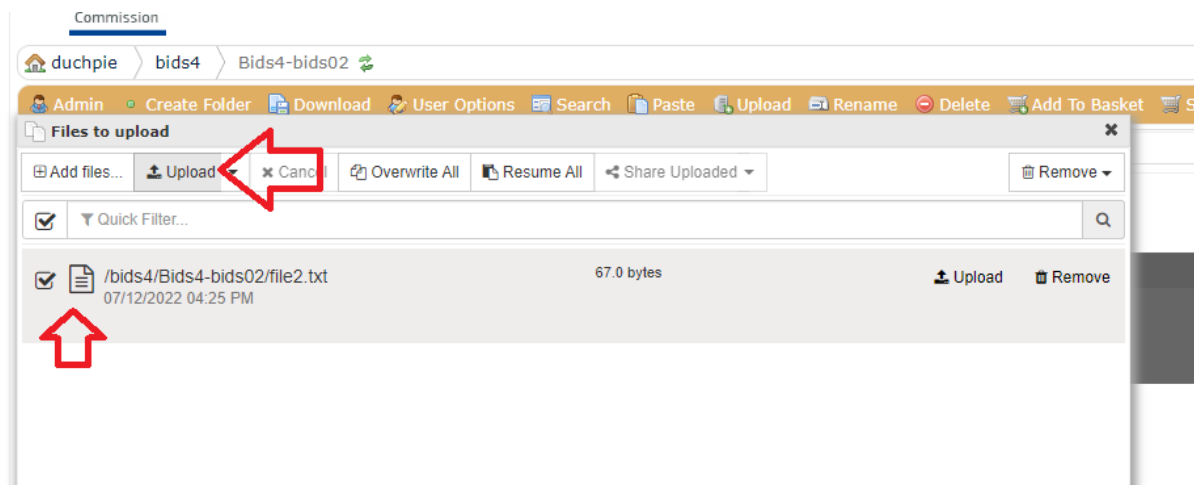
16. Navigate back to the new user's sub-folder.



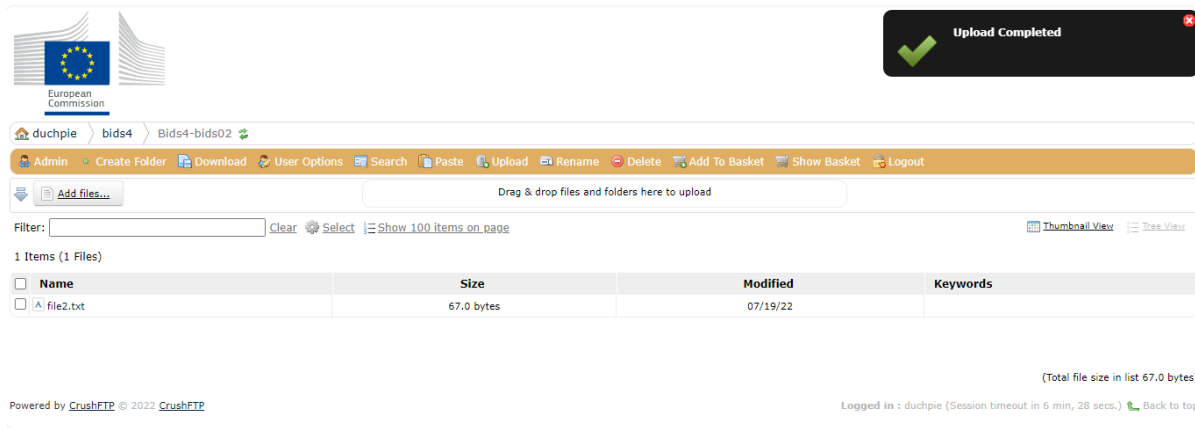
17. Click the 'Add files' button and navigate to file you wish to upload to the sFTP folder.



18. Select and click the 'Upload' button.



An 'Upload Completed' message is displayed and the file is located in the user's sub-folder.



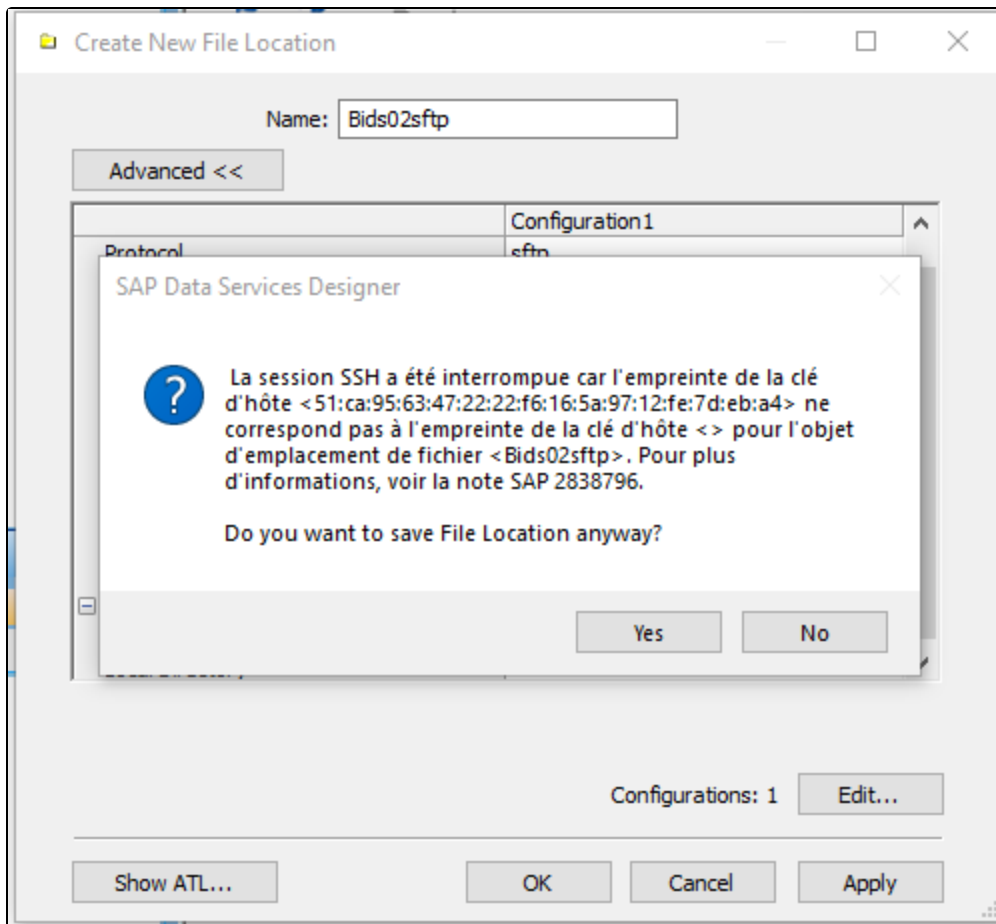
19. Log out of the EU Send Web Interface.

Designer - File Location

In the SAP BODS, **EU Send sFTP** is instantiated as a File Location object, that will host file objects made available to the SAP BODS platform as data sources. The EU Send sFTP service limits will apply in all circumstances to the file objects it hosts.

1. Log into the SAP Data Services Designer.
2. In the Format tab of the Local Object Library, create a new File location object.
3. Select sftp as the connection protocol.
4. Enter the EU Send host (eusend-sftp-acc.eusfx.ec.europa.eu for EU Send Acceptance environment)
5. Enter port (22222)
6. Enter EU Send hostkey fingerprint

(51:ca:95:63:47:22:22:f6:16:5a:97:12:fe:7d:eb:a4)



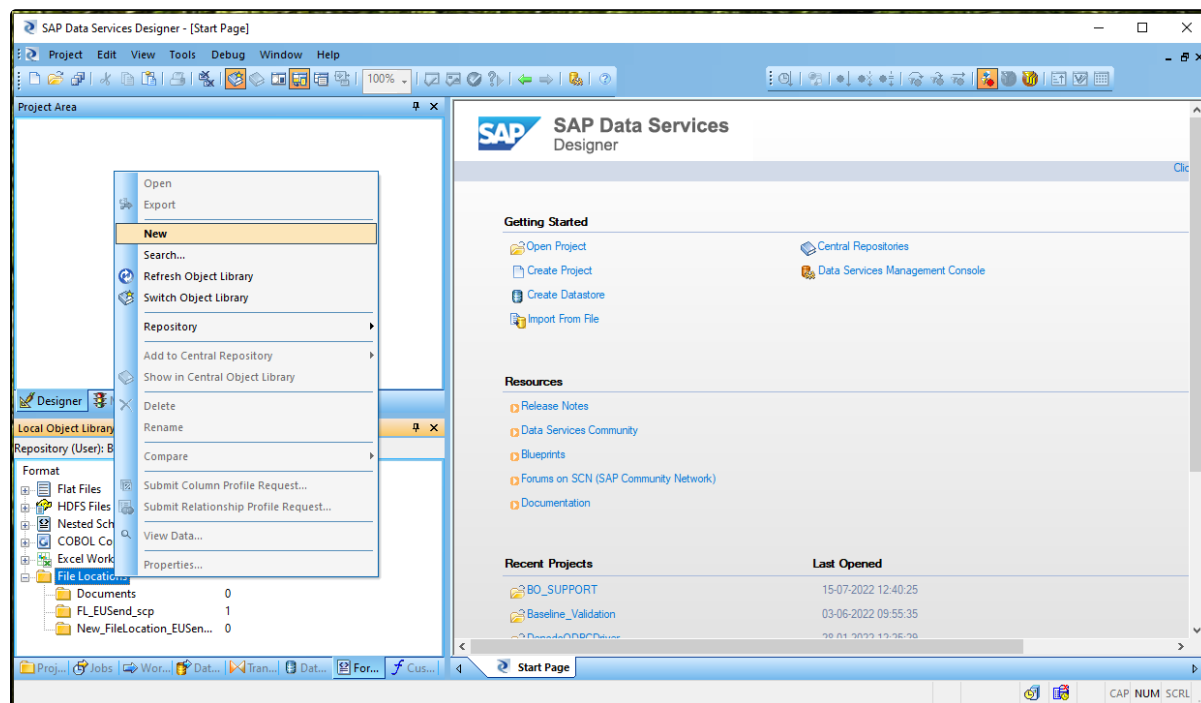
Hostkey value

This value can either be communicated by a request from the EU Send team or deduced from an initial error in the creation of this File Object.

If left blank on the first attempt to create the file object, the correct value is displayed in the error message. This instruction set will demonstrate this occurrence

7. Select Public Key as Authorization Type.
8. Enter user name (Bids-bids02)
9. Enter file path to private key in SSH Authorization Private Key File Path (/ec/<environment>/server/home/bods/keys/).
10. Enter the key passphrase in the SSH Authorization Private Key Passphrase.
11. Enter the file path to the public key in SSH Authorization Public Key File Path (/ec/<environment>/server/home/bods/keys/).
12. Enter the user's folder in the File System - Remote Directory using the syntax /<sub-folder name>/ , as example (/Bids4-bids02/).
13. Click 'Apply'.
14. Note the key fingerprint displayed, in the case for the EU Send Acceptance environment, 51:ca:95:63:47:22:22:f6:16:5a:97:12:fe:7d:eb:a4 is listed in the error message.
15. Click the 'Yes' button.
16. Enter the EU Send's host key fingerprint

51:ca:95:63:47:22:22:f6:16:5a:97:12:fe:7d:eb:a4
(This value is hashed in MD5)



SAP Data Services: How to connect SUMMA DWH with SAP Data Services



Homepage

Page Topics

- [Overview of SUMMA DWH with SAP Data Services](#)
- [What are the architecture components?](#)
- [Self-service - How to configure an Adapter to connect to SUMMA DWH](#)
 - [Create an SAP adapter](#)

Related Links

- [General FAQ](#)
- [General Recommendations](#)
- [How to connect EUSend with SAP Data Services](#)
- [How to use PrivX with SAP Data Services](#)



SUMMA DWH info



The Commission decided to replace ABAC with a new, state-of-the-art financial system to modernize, rationalize and simplify the EU’s financial processes with SUMMA: a next generation corporate financial platform.

SUMMA will introduce a new era in Financial Management for the Commission's DGs and EU Agencies at beginning of 2024.

For more information, click [SUMMA](#).



- DIGIT offers direct connectivity to the SAP HANA database component of SUMMA DWH via SAP Data Services.
- DIGIT is responsible for assuring the functionality of connectivity components based on service levels compatibility of the DS platform.

SAP Recommendations

- SAP recommends to the customers to keep their SAP BI/DS Platforms and their reporting DB on a supported version according to the PAM.
- See SAP Notes [1600140](#) and [3208938](#) for SAP Data Services 4.x and SAP Information Steward 4.x compatibility with SAP HANA.

DIGIT Support

- DIGIT can only offer best effort support should SAP HANA service levels evolve independently of DIGIT's SAP DS Service level offering.
- DG BUDG is responsible for operation and maintenance of the SUMMA DWH and related client applications /packages. This also includes user accounts and their corresponding rights.

Overview of SUMMA DWH with SAP Data Services

SAP Data Services offers two connector class objects to access directly an SAP HANA database (SUMMA DWH) as a *Datastore*.

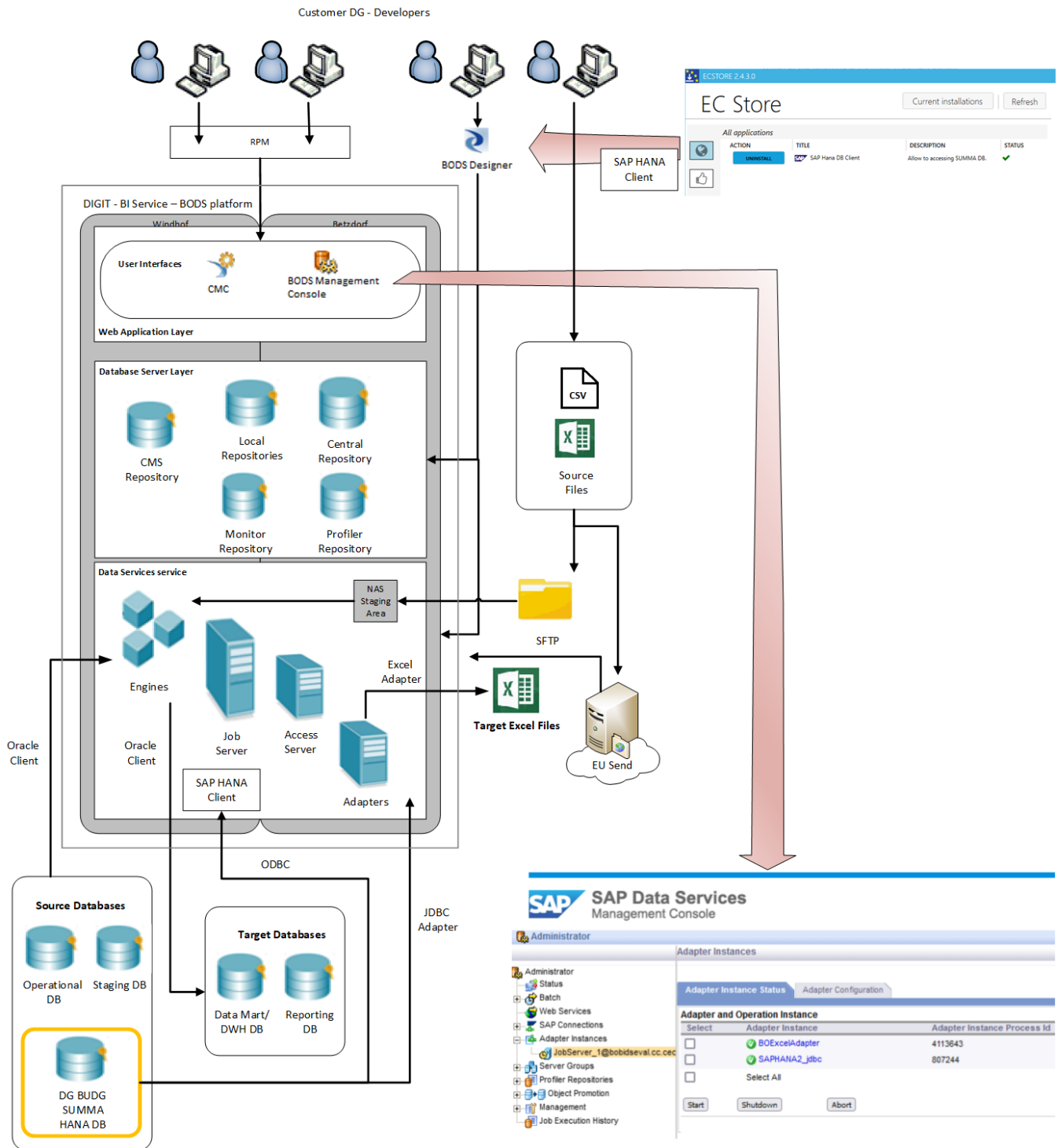
The first is the *Datastore* type: **Database**.

- requires the installation of the SAP HANA Client on the server and client workstations/laptops.
 - is based on ODBC, the driver via the configuration of the Data Services ODBC functionality.
 - DG BUDG is responsible for making available to the BI service an SAP HANA client for the Linux server and a package that is available to all users in the EC Store.
 - The BI service must configure the job server ODBC configuration files with our client's connection account information (user account, password, hostname and port number).
- brings the standard database type related functionality to the datastore, allowing importing of tables and views as source objects, along with the reconciliation of these in jobs and the preview of data.

The second is the *Datastore* type: **Adapter**.

- requires the use of the JDBC Adapter functionality. It adopts the self-service approach, as it can be configured entirely by users in the Data Services Management Console.
- brings the basic database type related functionality to the datastore, allowing importing of tables and views as source objects.

What are the architecture components?

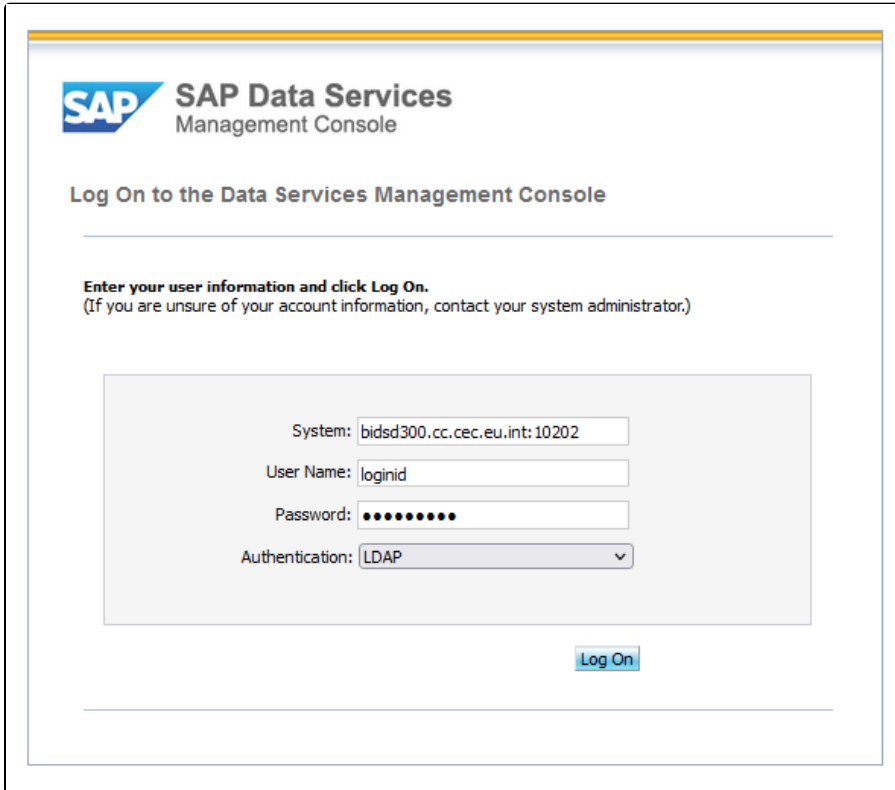


Self-service - How to configure an Adapter to connect to SUMMA DWH

Create an SAP adapter

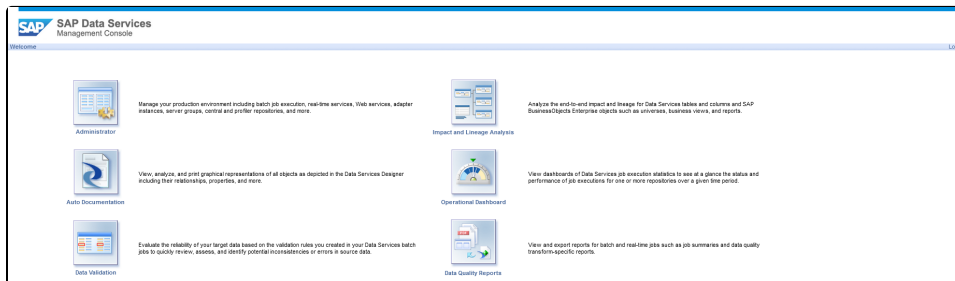
Users can connect to SUMMA DWH via SAP Data Services to create the Adapter (JDBCAdapter) themselves without intervention from DIGIT.

1. Log into the Management Console using your account.



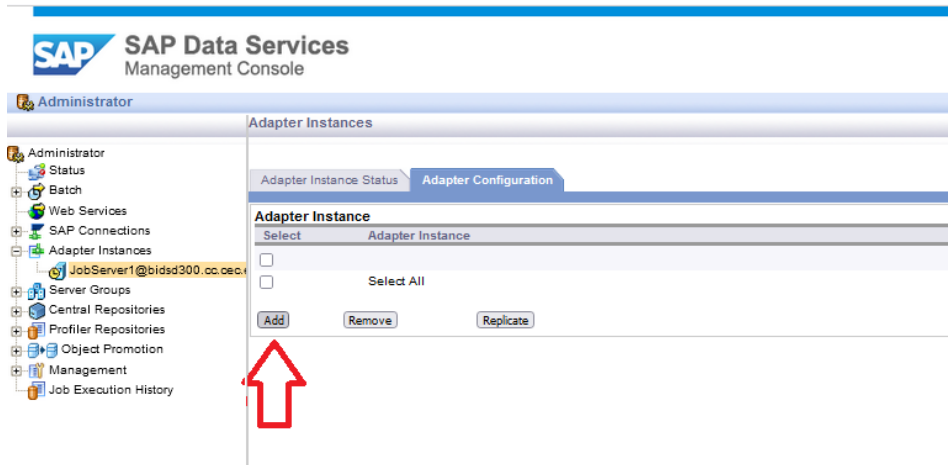
The screenshot shows the SAP Data Services Management Console login interface. At the top, the SAP logo is followed by the text 'SAP Data Services Management Console'. Below this, the heading 'Log On to the Data Services Management Console' is displayed. A message instructs the user to 'Enter your user information and click Log On.' and provides a note for users unsure of their account information. The login form contains four fields: 'System' with the value 'bidsd300.cc.cec.eu.int:10202', 'User Name' with 'loginid', 'Password' masked with dots, and 'Authentication' set to 'LDAP'. A 'Log On' button is positioned at the bottom right of the form.

2. Click on the 'Administrator' button.

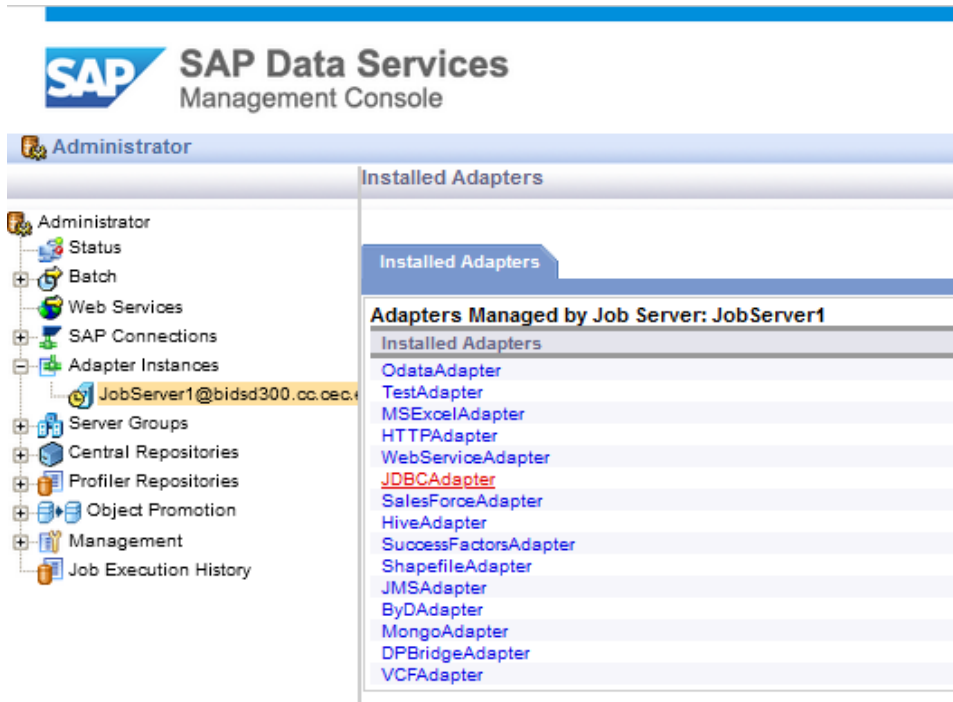


3. Navigate to the Adapter Instances and locate the 'Job server entry'¹.
4. Select the 'Adapter Configuration' tab.

5. Click the 'Add' button.



6. Select the 'JDBCAdapter' entry.



7. Configure the required fields of the Adapter instance with the following information:

The screenshot shows the 'Adapter Configuration' dialog in SAP Studio. The 'Adapter instance startup configuration' tab is selected. Red arrows indicate the following fields to be configured:

- Adapter Instance Name:** Must be unique. Becomes part of adapter's runtime definition.
- Use SSL Protocol:** Checked. Applies if adapter accesses real-time services.
- Classpath:** Contains a long file path. The -classpath Java parameter value when adapter starts.

Below the startup configuration, the 'JDBC Adapter JDBC connection and pushdown capability configuration' tab is visible, showing fields for 'JDBC driver class name', 'JDBC driver url', 'User', and 'Password'.

Adapter Instance Fields	Settings
Adapter Instance Name	Use a significant name such as SAPHANA2_JDBC .
Use SSL protocol	Set to False
Location of the JDBC jar file	Append the location of the JDBC jar file to the classpath value: <code>"/ec/{dev test prod}/server/bods/home/<user_bidsd999>/bods/sap_bobj/enterprise_xi40/java/lib/bundles/ngdbc.jar;"</code> <code>{dev test prod}</code> depends on the SAP DS platform's environment. For example: - Based on the image above with the system provided values, we see dev, so a development environment, with the <code>user_bidsd999</code> showing bidsd300. - We place <code>/ec/dev/server/bods/home/bidsd300/bods/sap_bobj/enterprise_xi40/java/lib/bundles/ngdbc.jar;</code> in the <i>Classpath</i> field. Do not forget the semi-colon ";" at the end of the line!
JDBC driver class name field	Enter com.sap.db.jdbc.Driver in the <i>JDBC driver class name</i> field.
JDBC driver url field.	Enter the following jdbc:sap://ciabatta.cc.cec.eu.int:34015/?databaseName=DT2 or: <code>jdbc:sap://<HANA Sservername>.cc.cec.eu.int:<HANA port number>/?databaseName=<database name></code> in the <i>JDBC driver url</i> field.
User field.	Enter the username in the <i>User</i> field.
Password field.	Enter the password in the <i>Password</i> field.

8. Skip the parameters that follow the JDBC Connection Information section.

9. Click the 'Apply' button at the bottom of tab window to save the adapter instance configuration.

[illegible]

JDBC Pushdown Capability

Support Simple Expression in GroupBy	Yes	Support pushdown of simple expression in GroupBy
Support Simple Expression in InnerJoin	Yes	Support pushdown of simple expression in InnerJoin
Support Simple Expression in OrderBy	Yes	Support pushdown of simple expression in OrderBy
Support Simple Expression in OuterJoin	Yes	Support pushdown of simple expression in OuterJoin
Support Simple Expression in Projection	Yes	Support pushdown of simple expression in Projection
Support Nested Expression In GroupBy	Yes	Support pushdown of nested expression in GroupBy
Support Nested Expression In InnerJoin	Yes	Support pushdown of nested expression in InnerJoin
Support Nested Expression In OrderBy	Yes	Support pushdown of nested expression in OrderBy
Support Nested Expression In OuterJoin	Yes	Support pushdown of nested expression in OuterJoin
Support Nested Expression in Projection	Yes	Support pushdown of nested expression in Projection
Support Nested Expression in Where Clause	Yes	Support pushdown of nested expression in Where Clause
Support Or Condition	Yes	Support pushdown of Or Condition
Support Nested Function In GroupBy	Yes	Support pushdown of nested function In GroupBy
Support Nested Function In InnerJoin	Yes	Support pushdown of nested function In InnerJoin
Support Nested Function In OrderBy	Yes	Support pushdown of nested function In OrderBy
Support Nested Function In OuterJoin	Yes	Support pushdown of nested function In OuterJoin
Support Nested Function In Projection	Yes	Support pushdown of nested function In Projection
Support Nested Function In WhereClause	Yes	Support pushdown of nested function In WhereClause
Support Functions	Yes	Support pushdown of functions

JDBC Math Function Support

abs	Yes	Returns the absolute value of the input number
ceil	Yes	Returns the smallest integer value greater than or equal to the input number
floor	Yes	Returns the largest integer value less than or equal to the input number
round	Yes	Returns the input number rounded to the specified number of decimal places to the right of the decimal point
trunc	Yes	Returns the input number truncated to the specified number of decimal places to the right of the decimal point
sqrt	Yes	Returns the square root of the input number
log	Yes	Returns the base-10 logarithm of the given numeric expression
ln	Yes	Returns the natural logarithm of the given numeric expression
power	Yes	Returns the value of the given expression to the specified power
mod	Yes	Returns the remainder when one number is divided by another

JDBC String Function Support

lower	Yes	Converts the input string to lowercase
upper	Yes	Converts the input string to upper case
rtrim_blanks	Yes	Returns the input string with the blanks (space characters) on the right removed
ltrim_blanks	Yes	Returns the input string with the blanks (space characters) on the left removed
length	Yes	Returns the length of the input string
substr	Yes	Returns the portion of the input string specified by the offset and length
soundex	Yes	Returns the soundex encoding of the input string

JDBC Aggregate Function Support

avg	Yes	Calculates the average of given set of values
count	Yes	Counts the number of values in a table column
count_distinct	Yes	Counts the number of distinct non-null values in a table column
max	Yes	Returns the maximum value from a list
min	Yes	Returns the minimum value from a list
sum	Yes	Calculates the sum of a given set of values

JDBC Date Function Support

week_in_year	Yes	Returns the week number relative to the year for the input date
month	Yes	Returns the month number of the input date
quarter	No	Returns the number of the quarter for the input date
year	Yes	Returns the year number of the input date
day_in_month	Yes	Returns the day number relative to the month for the input date
day_in_year	Yes	Returns the day number relative to the year for the input date

JDBC Miscellaneous

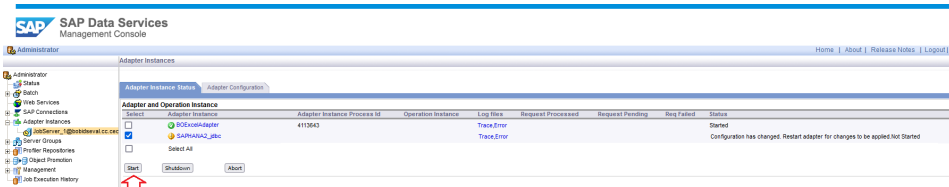
ifthenelse	No	Computes the expres A. If A evaluates to TRUE, returns B. Otherwise, returns C.
nvl	Yes	Replaces Input with Replacement if Input is NULL



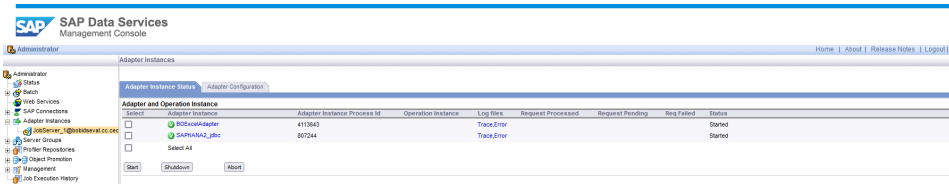
10. Click on the 'Adapter Instance Status' tab.

11. Check the checkbox under the Select column for the **SAPHANA2_JDBC** instance just created.

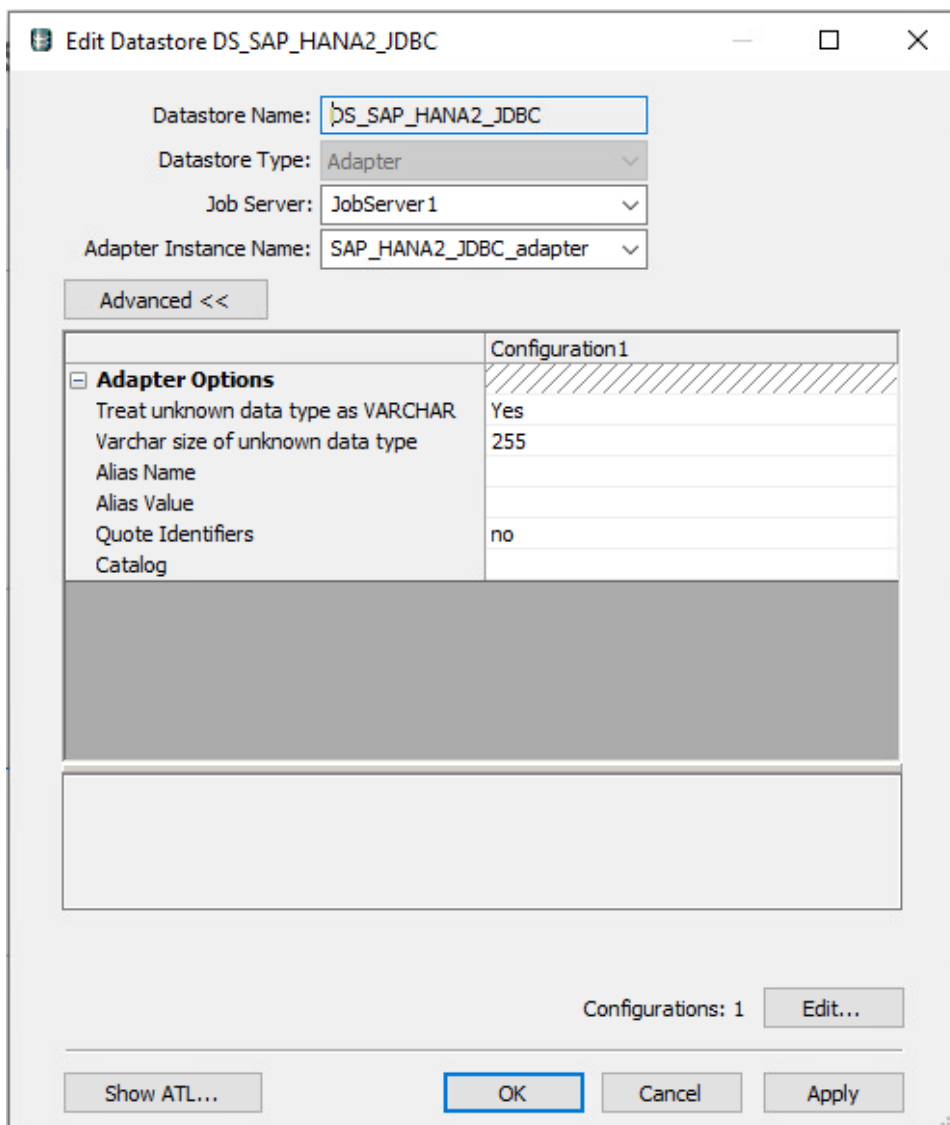
12. Click the 'Start' button to launch the adapter process.



13. Verify that the adapter instance is started without errors.



14. Check that you can now create adapter data sources in the SAP DS Designer.



SAP Data Services: How to use PrivX with SAP Data Services



Homepage

Page Topics

- [Pre-requisites of Oracle Database COP with SAP Data Services](#)
- [Overview of COP with SAP Data Services](#)
- [Architecture of Oracle Database COP used for SAP Data Services](#)
- [How to connect to Oracle Database COP Repositories with SAP Data Services Designer?](#)

Related Links

- [General FAQ](#)
- [General Recommendations](#)
- [How to connect EUSend with SAP Data Services](#)
- [How to connect SUMMA DWH with SAP Data Services](#)



Oracle COP Info

The European Commission has decided to replace legacy Oracle DB with Cloud-on-Premise (COP).

For more information about Oracle Database Cloud On-Premise (COP) service, please refer to the appropriate [User Guide](#).

Pre-requisites of Oracle Database COP with SAP Data Services

To connect to the SAP Data Services Designer and access the SAP DS Repositories (Local, Central, and Profiling), users must have repository access and the appropriate permissions defined in the Central Management Console (CMC) of the SAP BO Portal.

It also requires that the Group/User has requested access to the "flow" for Oracle in PrivX: See [JASSPR](#).

Install PAM PrivX Gateway in the computer from the Software Center (softwarecenter:SoftwareID=ScopelId_2FDB684F-0DB7-4DA6-AF92-C1DA1E3DFC65/Application_01c68017-8416-4cb3-96f1-145852c4d9ce):

Software Center

European Commission

Applications

Updates 1

Operating Systems

Installation status

Device compliance

Options

All Required Featured

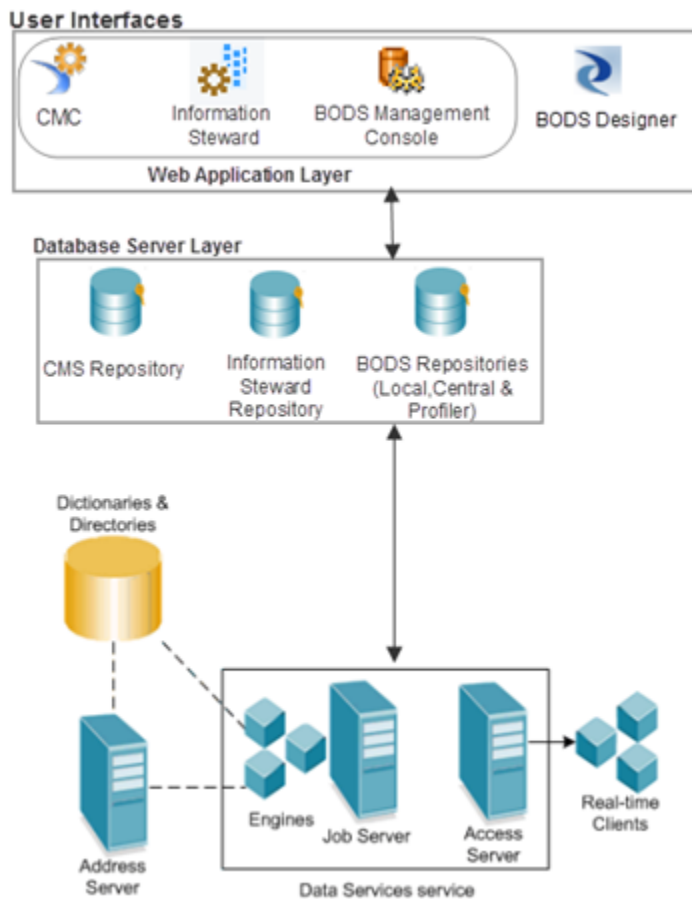
Filter: All Sort by: Most recent

Name	Publisher	Version
PAM PrivX Gateway PROD	EC	1.2a

To access to the Local repositories must be provided the required LDAP groups that must be added (*LDAP Group and Environment*). Flow needs to be open by CoE.

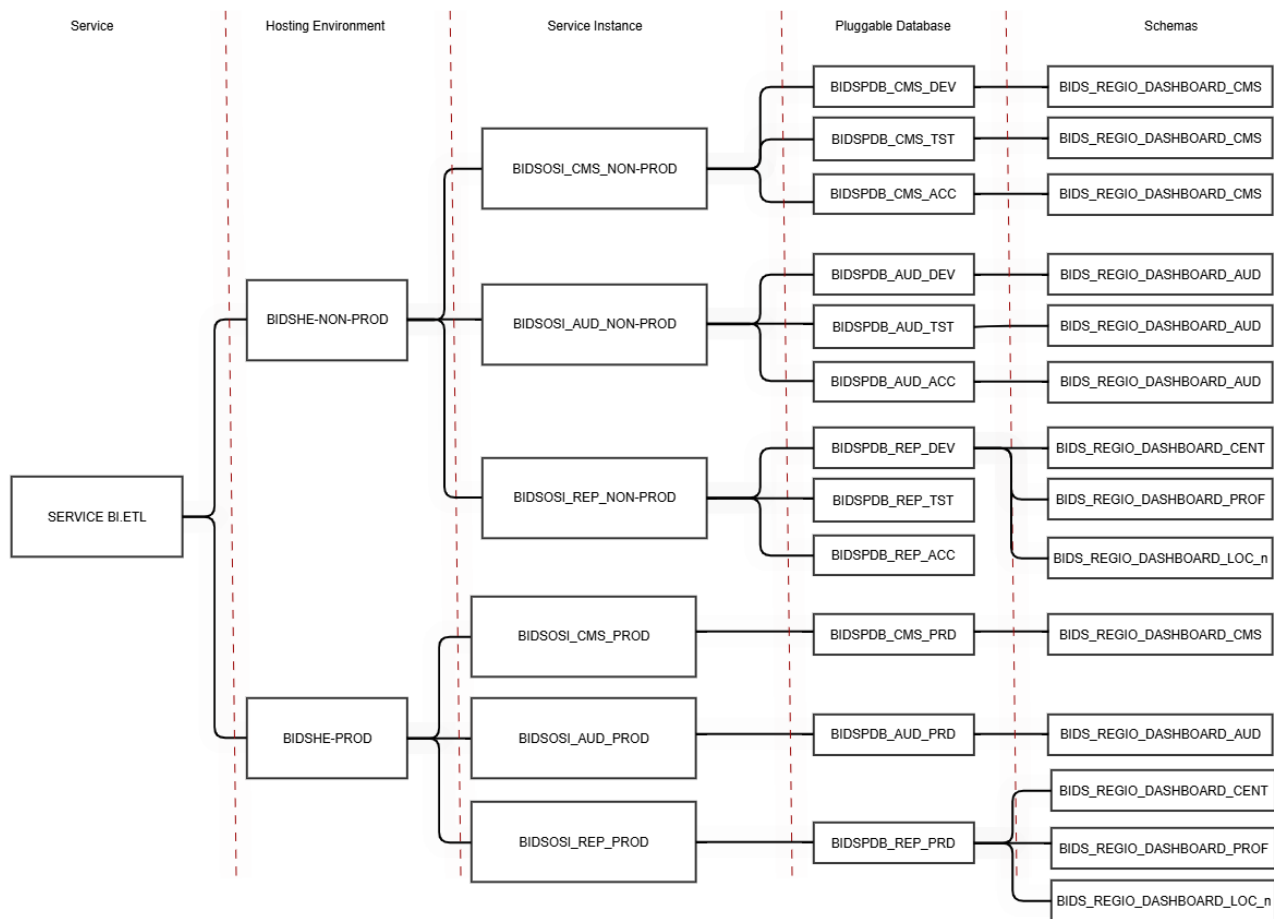
Overview of COP with SAP Data Services

The SAP Data Services Designer client tool enables users to develop ETL over local repositories.



Architecture of Oracle Database COP used for SAP Data Services

The following architecture has been defined for SAP Data Services:



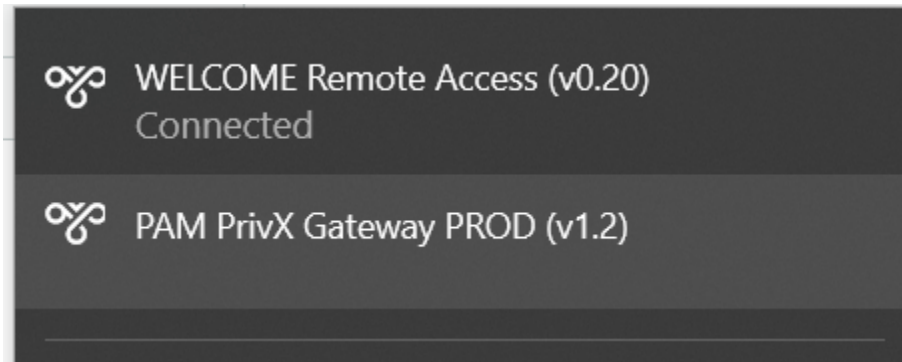
i For this documentation and for SAP DS Designer we must focus on the PDBs *BIDSPDB_REP_XXX* where XXX is the environment (PRD for production, ACC for Acceptance, TST for Test and DEV for development).

How to connect to Oracle Database COP Repositories with SAP Data Services Designer?

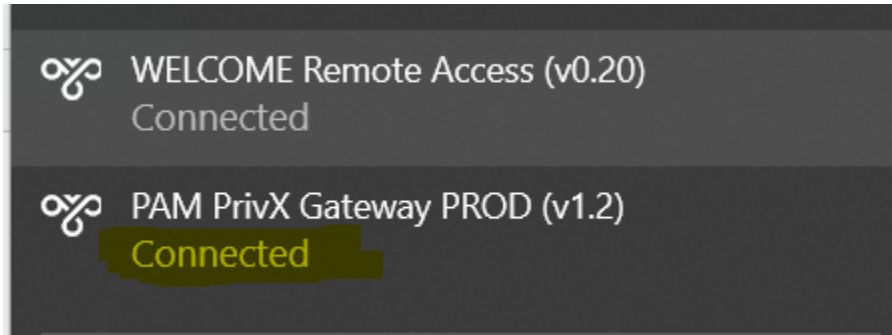
i Prerequisites

Before starting the SAP DS Designer, you must connect to the PrivX VPN and then to the PrivX network where the Repository is located.

1. Connect to the PAM PrivX VPN:



after connect:



2. Connect to [Home | PrivX](#).
3. Go to Connections → Network Targets → [Network Targets | PrivX](#)
4. Select which network you want to connect if PROD or NON-PROD (DEV, ACC, TST):

BIDSOSI_REP_NON-PROD.13F1:VM	10.130.9.144 (TCP 1521)
BIDSOSI_REP_PROD.13F0:VM	10.130.9.138 (TCP 1521) 10.130.9.146 (TCP 1521)

After connect:



The network target session will remain open as long as this page is kept open. Closing the page will terminate the session.

Name	BIDSOSI_REP_NON-PROD.13F1:VM
Status	Connected
Established	2025-09-01 13:26:11
Duration	0:01
Exclusive Access	Disabled
Source NAT	Enabled
Client IP	10.148.250.170
Destinations	10.130.9.144 (TCP 1521)

When connected to PrivX VPN and Destination Network, you can open SAP Data Services and connect to your local repositories as you use the SAP Data Services Designer.



Disconnect from PrivX

Don't forget to disconnect from PrivX Network and VPN.

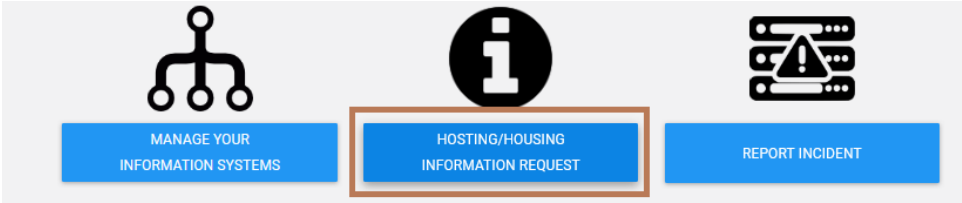
SAP Data Services: Support & Useful Links



Homepage

Page Topics	Related Links
- Support - Focus of Support - Roles & Responsibilities - DIGIT Responsibilities - Customer Responsibilities	FAQ

Support

Link	Description
SAP Data Services Service Catalogue	SAP Data Services Service Catalogue
What is new?	Check the Change Log to see what's new and track past changes
DIGIT Price List	DIGIT Price List for Infrastructure Services
Contact the SAP Data Services team	Raise a Request for Information 
Incident Management	Resolve issues efficiently and in a way that minimizes the impact from incidents
Incident Management (Generic)	Prime objectives of Incident Management

JASSPR Tutorial Videos	• Incidents • Request for Change (RFC) • Non Standard Request for Change (RFC) • Request for Information (RFI) • Request for Service (RFC)
JASSPR	Access to JASSPR • Access from EC networks (European Commission and Executive Agencies): https://intragate.ec.europa.eu/jasspr • Access from TESTA network (EU Institutions and Agencies): https://webgate.ec.testa.eu/jasspr
EU Login	Authenticate using EULogin
JASSPR User Guide	Homepage for JASSPR User Guide
Dynatra ce Monitori ng - SAP DS Dashbo ard	Dashboard in Dynatrace to monitor SAP BO DS Hosts with CPU, Memory and Disk.

Focus of Support

- 24/7 Support for the PRD environment.
- 12/5 Support for other environments
- Daily backup of data
- Restoration of data
- Upgrades and patches
- Versions
- Security Features
- Creating an instance.
- Modifying an instance.
- Decommissioning an instance.

Roles & Responsibilities

DIGIT Responsibilities

DIGIT RESPONSIBILITIES	DESCRIPTION
Deployment architectures	Providing two deployment architectures: • Stand-alone VM for DEV & TEST environments. • Clustered VMs for PRD environments.

Backup & Restoration	• Daily backup of file systems (SAN & NAS) • Retention of four weeks. • Restoration of customer database upon request (not including default service catalogue actions).
Upgrades	• DIGIT: ○ Validates upgrades ○ Manages the upgrade process
Patches	Quick and efficient installation of OS security or technical patches.
Security	DIGIT performs and supports the required configuration steps regarding the authentication.
Monitoring	Monitoring the availability of the Data Services instance.
Support	• 24/7 support for Production service instances • 12/5 support for other environments  Refer to the Support section of Information System Hosting Services to find out how to get help and the point of contact. The Hosting Services follow a common model to manage incidents. Consult Incident Management for more information.
Instances	• Creation of an instance • Modification of an instance • Providing access to an instance • Decommissioning an instance • DIGIT performs the decommission but the customer is responsible for their data after the decommissioning process.

Customer Responsibilities

CUSTOMER RESPONSIBILITY	DESCRIPTION
Security	• Verifying that users have been granted sufficient rights to perform the requested actions upon the specified object. • Safeguarding the history of customer application artefacts. • Regular password modification.
Performance	Issues not directly related to the service provided by DIGIT: • Performance issues not related to the server • Application tuning for performance improvement

Requests	The customer must make a request through JASSPR in order for DIGIT to fulfil the request: • Creating an instance • Modifying an instance • Decommissioning an instance The customer is responsible for their application data after decommissioning.  This includes data in the Staging Area, as the Staging Area will be decommissioned with the instance. All residing data will be lost.
Consulting activities	• The customer must use external consultancy services if required. • This service is not available in DIGIT currently.
Support	For example: • Customer application support • Debugging and fixing application errors.  Refer to the Support section of Information System Hosting Services to find out how to get help and the point of contact. The Hosting Services follow a common model to manage incidents. Consult Incident Management for more information.
IS Risk Assessment	• It is the responsibility of the customer to perform a risk assessment of the Information System (IS) and to select suitable services.  DIGIT provides support to the customer in order to help them make optimal choices and identify which services are best suited for an information system.